Racing to the Bottom: Performance Thresholds, Venue Advantage, and the Market for Downhill Marathons*

Adam Looney†

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Abstract

Many institutions allocate scarce opportunities or prizes to applicants who surpass a performance threshold. When candidates can select favorable conditions in which their performance is measured, their behavior can misallocate both opportunities and the venues where performance is measured. I study this problem through the example of the Boston Marathon's qualifying system, where runners must submit a qualifying time but can choose any certified marathon. Using 17.2 million finish times and an AKM decomposition of performances into runner and course effects, I show that certain courses, often steeply downhill, produce time advantages of nearly 20 minutes at the qualifying margin. Marginal runners sort to fast courses, finish times at fast courses bunch at qualifying standards, and a number of fast, downhill marathons entered the market. As a result, among linked Boston entrants who can be classified, the share whose qualification depends on their course roughly doubled, from 14.5% in 2004 to 29.1% in 2025; adjusting qualifying times for course speed would change 11.9% of who qualifies, holding the number of qualifiers fixed, and those newly qualifying are predicted to run Boston about 4.9 minutes faster than those they replace. Beginning in 2027, Boston will penalize some steeply downhill courses; the announced rule eliminates only about 6.3% of the disagreement with the common-course benchmark.

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†Marriner S. Eccles Institute, David Eccles School of Business, University of Utah. Email: adam.looney@eccles.utah.edu.

Many institutions allocate scarce opportunities using a common performance threshold such as admission for students above a test-score or GPA cutoff, promotion for workers who hit a production target, or selection into competitions for athletes who meet a qualifying standard. These systems may appear meritocratic because everyone faces the same bar. But when candidates can choose the environment in which performance is measured, equal thresholds need not imply equal standards. If students can choose classes with different grading rigor or seek testing accommodations, teachers can select classrooms with favorable student composition, or workers can pursue assignments where target attainment is easier, then a common threshold confounds ability with testing environment. And if a screening rule rewards favorable environments, a market for evaluation environments may arise.

I examine this issue in the context of amateur runners pursuing scarce qualifying slots at the Boston Marathon. To qualify, runners must achieve a qualifying time—based on age and gender—at a recent marathon. The qualifying standard (the BQ for "Boston qualifier") is public, the set of qualifying marathon courses is observable, runners freely choose among them, and marathon venues enter, grow, and market themselves partly around the rule. As marathon running became more popular and qualifying for Boston became a more celebrated achievement, the number of applicants increased, driving the qualifying standard progressively tighter. The qualifying threshold in 2026 was 15 minutes faster for most runners than in 2011. Then in 2025 the BAA went a step further, announcing that beginning with the 2027 race it will apply penalties to qualifying times from steeply downhill courses out of an apparent concern that some runners were using such courses to gain undue advantage.

To understand the role of course differences and venue selection in the market for Boston qualification, I gather data on 17.2 million public finish times at over 1,000 marathon courses, and use that information to link runners and their times across races. I then estimate an AKM-style decomposition of log finish times into runner and course effects (Abowd, Kramarz and Margolis, 1999; Card, Heining and Kline, 2013). If the same runner is systematically faster at one course than another, after accounting for age and race-day weather, the repeat-runner difference identifies a course advantage. The key assumptions of the AKM approach are broadly supported in this setting, with one exception: marginal runners pursuing Boston qualification clearly 'peak'—expend selective, targeted effort—at qualifying courses. I propose a correction that recovers course effects net of this behavior, exploiting the directional asymmetry of performances before and after the Boston Marathon.

For runners at the qualifying threshold, the gap between the fastest and slowest courses exceeds 20 minutes. The course effects are strongly predictive of subsequent performance. Runners who qualify for Boston on a course estimated to be 1% faster subsequently run Boston about 0.85% slower than their peers. Moreover, the course effects are closely related to observable characteristics of the marathons themselves: gross elevation change, in particular, is the dominant physical predictor of how fast a course is.

The screening distortion is large and growing. Two statistics summarize it, and they measure different things. The first is course dependence. The share of apparent-qualifier performances whose posted-standard qualification depends on the course rises from about 9.4% in the Boston 2004 cohort to about 17.4% in the Boston 2025 cohort; among linked Boston entrants who can be classified (about 40.0% of the recent field), the share with no observed performance that would clear the effective cutoff on a Boston-equivalent course rises from 14.5% in 2004 to 28.5% in 2019 and 29.1% in 2025. Course dependence means these entrants' qualification rests on where they ran; it does not by itself say who would lose qualification under a corrected rule. The second statistic, displacement, answers that ranking question directly: holding the number of qualifiers fixed, a venue-adjusted standard would change about 11.9% of who qualifies, shifting qualifications from fast courses to slower ones, and those newly qualifying are predicted to run Boston about 4.9 minutes faster, on average, than those they replace.

Part of the increase comes from runners deliberately choosing faster courses when pursuing a BQ. Runners near the qualifying threshold select fast courses at disproportionate rates and that selection has increased over time. Runners bunch at the Boston qualifying standard, and this bunching is concentrated at fast courses. When the threshold declines, the bunching at fast courses follows the threshold.

Fast courses also became more common. The number of courses at least five minutes faster than Boston at the M18–34 standard increased 1.9-fold (from 23 to 44 between 2004 and 2025). Purpose-built Boston-qualifying products, like the REVEL series and the "BQ.2" races, entered on a timeline aligned with announced tightenings. These marathons often market themselves explicitly as qualifying venues or are highlighted on websites offering advice to would-be Boston qualifiers. Aided by these entrants, the share of US finishers at these courses nearly doubled (from 5.7% to 11.1%). In a decomposition of the misallocation rate, the venue-supply channel contributes 5.5 percentage points by the Boston 2025 cohort (3.0–6.2 across counterfactual baselines and orderings), on par with runner-side sorting (7.2 points, 6.0–9.6). The result was a growing market for races offering unusually favorable qualifying conditions.

This paper relates to several literatures: candidates gaming common measures (Sabot and Wakeman-Linn, 1991; Jacob, 2005; Neal and Schanzenbach, 2010; Lazear, 2006), AKM-style measurement and its validation (Abowd, Kramarz and Margolis, 1999; Card, Heining and Kline, 2013; Chetty, Friedman and Rockoff, 2014; Rothstein, 2010), and markets for favorable evaluation (Bolton, Freixas and Shapiro, 2012). The closest empirical relative is Dranove et al. (2003), who show that published report cards induced providers to select patients.

The policy section compares four approaches: the current common cutoff, the BAA's one-sided net-drop penalties, a transparent symmetric physical adjustment, and a behavior-adjusted course-effect handicap. Estimated course effects are most reliable for courses with long histories; a physical formula can cover new or sparsely observed courses, but it should credit difficult courses as well as penalize favorable ones. The comparison is run on U.S. road courses and evaluated out of sample; trail races inform the measurement but sit outside the qualifying market the rule governs.

Every fall, thousands of runners a few minutes too slow for the Boston Marathon travel to races built to fix that. Some descend four or five thousand feet through mountain canyons. Others are flat, small, and paced to the second. One series puts the product in its name: the "Last Chance BQ.2" marathons sell a Boston-qualifying time, a "BQ." The strategy works: the share of Boston's field qualifying on courses that drop more steeply than Boston itself has more than doubled since 2004, from 9 to 21%.

The Boston Marathon is the world's oldest annual marathon and one of its most prestigious. Unlike most large-city marathons, Boston relies on time-based qualification rather than a lottery. The threshold is high-stakes, clearly stated, and publicly known; as BAA President Jack Fleming puts it, runners "earn a spot on the starting line" (Boston Athletic Association, 2025*b*).

Regular increases in applications have made Boston substantially more selective. For men aged 18–34 the posted standard tightened from 3:10 (through Boston 2012) to 3:05 (2013–2019), then 3:00 (2020–2025), and now 2:55 (2026–27). On February 16, 2011, the BAA announced both the rolling fastest-first registration process (effective for the 2012 race) and the five-minute tightening of all standards (effective for 2013); the September 2018 and September 2024 announcements followed the same pattern.[1]

The qualifying system applies the same time cutoff across all certified marathons regardless of course difficulty: a 3:00 at the steeply downhill REVEL Big Bear and a 3:00 at the hilly Marine Corps Marathon are treated identically, though Section 1 estimates they reflect underlying performance differences of more than 20 minutes. In 2025 the BAA announced, for the 2027 race, net-elevation-drop penalties (5 minutes for 1,500–2,999 feet of net drop, 10 minutes for 3,000–5,999, and disqualification at 6,000+) (Boston Athletic Association, 2025*a*), the first departure from the common-cutoff rule. Section 4 evaluates how much of the venue advantage it actually corrects.

[1] In oversubscribed years the effective cutoff is the posted standard minus an admission buffer, unknown until registration closes, because Boston admits qualified applicants in order of their margin until the field fills. The posted standard is the target runners pursue and governs the behavioral analyses; the effective cutoff governs actual admission and enters the policy counterfactuals.

The data are a web-scraped panel of marathon results from race websites and aggregators (MarathonGuide.com, 2024): 17,206,046 finish-time observations at over 1,000 marathon courses in the US and 18 other countries, 2000–2026. A multi-stage entity-resolution procedure links runners across races by name, gender, and year of birth, erring toward conservative matching because false merges create spurious mobility links between courses while false splits only discard information. AKM estimation uses runners observed at least twice within their geographic region (11,142,687 observations on 2,631,055 regional runner identities at 1,162 courses); the 2.0 million runners observed on more than one course provide the cross-course comparisons that identify the course effects. Restricting to the highest-confidence identity matches leaves the course ranking essentially unchanged (Spearman rank correlation 0.95; Table B5). The selection and choice analyses additionally use a geocoded US-resident subpopulation of about 1.9 million runners. Appendix A details the entity resolution, sample construction, and time-variable selection, and Table A2 summarizes the samples and where each is used.

An apparent qualifier meets the posted standard; an effective qualifier clears the effective cutoff. A fast course is at least 2% faster than Boston on the AKM log scale (about four minutes at a 3:20 pace); very fast is at least 4%. "Near-BQ" means close to the runner's own standard; exact bandwidths vary by exercise and are collected, with each exercise's rationale, in Appendix A (Table A3).[2]

Observed performance combines four things a threshold cannot separate: the candidate's underlying ability, the systematic advantage of the environment they are evaluated in, the effort they bring, and luck. A principal who ranks candidates on that single number, against one common bar, faces three consequences (cf. Baker, 1992). A common threshold confounds person and place, so equally capable candidates qualify at different rates purely by where

[2] Two global conventions hold throughout. Eras are defined by the standard in force for the relevant Boston cohort (pre-2013 / 2013–2019 / 2020+ for cohort exhibits) or by the announcement cutovers (pre/post September 2011 and September 2018, when the first qualifying windows governed by each announcement opened; the 2011 announcement itself came in February) for exhibits that classify qualifying performances by regime; each exhibit states which. COVID-era observations are handled uniformly: the cancelled Boston 2020 cohort is shaded or omitted, and the atypical Boston 2021 field is suppressed from take-up calculations.

they are measured. Tightening the threshold amplifies the venue advantage, because as the bar moves into the thin upper tail of the ability distribution a fixed environmental edge pushes more candidates across it. And the strategic response concentrates among marginal candidates, who have the most to gain from a favorable venue.

The screening distortion arises whenever a common threshold is applied to heterogeneous venues, even absent strategic behavior. The market distortion is the endogenous response: candidates redirect venue choices, and venues themselves enter and grow, to exploit the advantage.

A prospective venue pays a fixed cost to enter, offers a systematic advantage, and earns margins on the candidates it attracts. Its customers are the candidates for whom the advantage has option value: those within reach of the standard only at a favorable venue. Each tightening leaves a dense band of candidates just outside the new standard for whom a favorable venue restores qualification; the return to entry therefore rises when the standard tightens, and venues built to supply the advantage should enter after announced tightenings. And because the rule rewards the advantage however produced, entrants will produce it in whatever form the rule does not police. Section 3 tests the timing prediction; Section 4 returns to the second prediction.

The policy analysis takes one normative position: a venue-adjusted standard should neutralize the course effect δ_c but not effort. Gravity is not merit; training to peak at a goal race is what the institution means to reward.

1 Measuring Course Advantage

I estimate a course-speed index on the full panel and then test, assumption by assumption, whether it should be believed.

I estimate

$$\ln T_{ict} = \alpha_i + \delta_c + \gamma_t + f(\text{age}_{it}, \text{male}_i) + g(W_{ct}) + \varepsilon_{ict}, \tag{1}$$

where T_{ict} is the finish time of runner i on course c in year t, α_i is a runner effect, δ_c a course effect, γ_t a year effect, f an age–gender polynomial, and g a quadratic in demeaned race-day temperature. Identities are defined within five geographic regions to limit false links; estimates are normalized to Boston ($\hat{\delta}_{\text{Boston}} = 0$) using about 24,000 bridge runners, shrunk by Empirical Bayes (noisy effects are pulled toward the mean so sparsely observed courses are not over-interpreted), and published for the 1057 courses with at least 50 observations. Throughout, $\hat{\delta}$ denotes this published, shrunk index unless a variant is named. Appendix B gives the full specification, the progressive R^2 decomposition, the shrinkage estimator, an algebraic invariance result (the fixed-field displacement of Section 2 is invariant to any common shift of the index, so only cross-course differential error can matter for the headline), and sample-restriction robustness.

Figure 1 plots the estimates. Purpose-built downhill courses run 4–9% faster than Boston; major city races run slower; for a runner at the qualifying margin, the difference between the fastest and slowest courses exceeds 20 minutes (the full published index is available as a CSV in the replication archive). Most courses are slower than Boston, so most BQ pursuit happens at a venue disadvantage, a fact Section 2 returns to when it measures the reallocation.

Physical predictors work in the expected direction, but most of their apparent power comes from trail and mountain races, where gross climbing alone has a cross-validated R^2 of 0.85. In the policy-relevant sample of 439 profiled U.S. road courses, the course-weighted out-of-sample R^2 rises from 0.07 for net drop to 0.33 for gross climbing and 0.47 for the full physical-and-weather formula. Even that formula explains only 0.12 of the variation weighted by where runners actually race. A symmetric physical formula is therefore a useful fallback for new courses, not a substitute for repeat-runner course effects. Nor is the drop in predictor fit a failure of the AKM index: re-estimating the model after excluding trails leaves the remaining course effects almost unchanged (Spearman $\rho = 0.999$; mean absolute change 0.2 minutes; profile coverage and detail in Appendix B). The median within-course year-to-

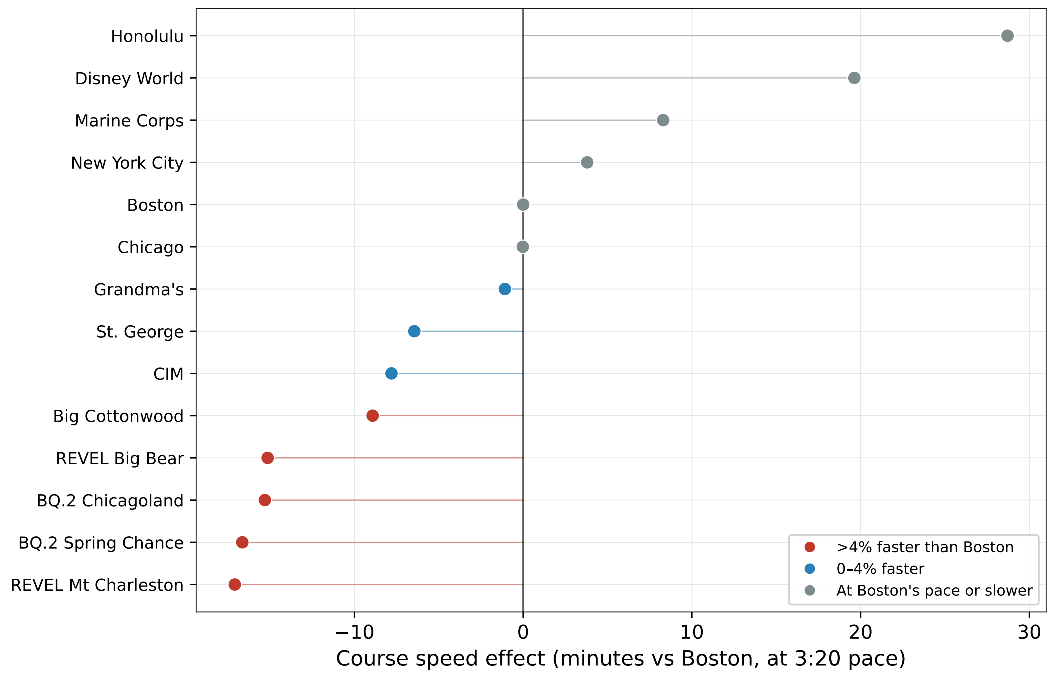

Figure 1: How much marathon courses differ. Recognizable courses in minutes relative to Boston at a 3:20 pace; negative values are faster than Boston.

year standard deviation of course means is 2.8 minutes, small against a cross-course range of more than 40 minutes (Appendix B). The most direct validation asks whether estimated course differences predict changes in the same runner's performance across courses; that comparison also tests the model's central assumption.

Equation (1) is an AKM-style decomposition with runner and course fixed effects, the same additive person-plus-environment structure as wage models (Abowd, Kramarz and Margolis, 1999; Card, Heining and Kline, 2013) and teacher value-added (Chetty, Friedman and Rockoff, 2014); repeated observations of the same runners across races identify systematic differences in course speed (Kline, 2024 and Bonhomme, Lamadon and Manresa, 2026 provide users' guides to the wage-model version). The estimates rest on three assumptions: additive separability, enough movement to separate person from place, and movement unrelated to transitory performance. In most applications these must be defended indirectly, because the environments are hard to characterize and the movers are few; here each can be tested directly. The first two hold. The third does not: marginal runners time and place their qualifying attempts strategically, and that behavior is what the rest of the paper studies.

Additive separability says a course shifts every runner's log time by the same amount. Most applications must assume it untested, because testing it requires observing the same person in many environments; here, millions of repeat-runner pairs test it directly. For each same-runner pair of races, the change in log finish time should equal the difference in the two courses' estimated effects, one for one. It does: regressing within-runner time changes on course-effect differences gives $\hat{\beta} = 0.933$ across all 10,934,525 cross-race pairs, and the Boston retest (pairing each Boston run with the same runner's prior marathons within 18 months, so that the index estimated elsewhere is tested out of sample at Boston itself) gives $\hat{\beta} = 0.851$ on 690,058 pairs (Figure 2; Appendix B, Table B1). Nor does the validation rest on Boston-destination pairs, the most campaign-selected population in the data: rerunning the retest with other destinations gives slopes of 0.99 (Chicago), 1.00 (New York), and 0.83 (Marine Corps) (Appendix Table B3), so the transfer relationship is not an artifact of the Boston-bound population alone.

The slopes are close to one. False entity matches among common names compare different people and push the slope above one; restricting to medium-frequency names with low-confidence links removed brings it to 0.98 overall and 0.93 into Boston (Appendix Table B2). Conversely, the slope into Boston sits below the slopes into the lottery-entry majors Chicago and New York: a performance peaked at the qualifying race does not transfer fully to the goal race. Additivity is therefore a good approximation, while its signed residuals identify the two biases the paper treats below.

The second assumption is enough movement to separate person from place. Here 1,985,267 runners are observed on more than one course, connecting the 1,162 courses, about 1,708 movers per course, so the mover graph is dense. I nevertheless avoid variance-decomposition claims, shrink noisy course effects toward Boston with Empirical Bayes, and report a conservative index that sets statistically imprecise effects to zero. Sampling error is too small to explain the policy results: leave-out estimates are nearly identical, a pure-noise placebo

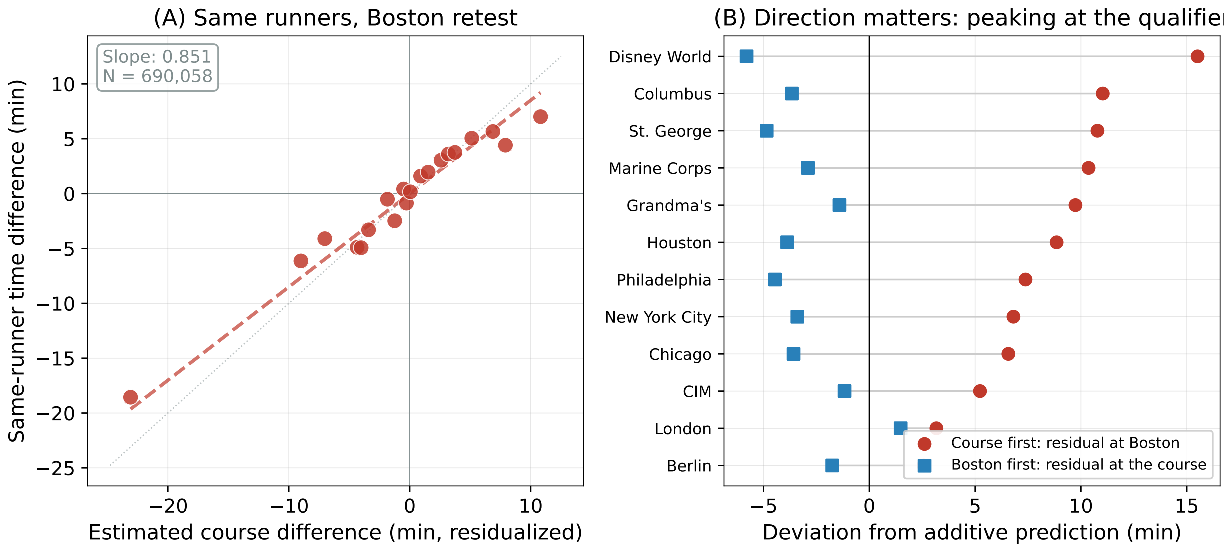

Figure 2: Why repeat runners identify course speed. Panel A: for the same runner, the time difference between two races should equal the estimated difference between their courses; points are pair-weighted quantile-bin means with age and gender partialled out of both axes, the dashed line is the pair-weighted regression slope, and the dotted line is the one-for-one benchmark. Panel B: runners are +7.9 minutes slower at Boston than the additive model predicts when the other course came first, versus -2.1 minutes when Boston is the origin. Systematic course differences should reverse sign when the order reverses; peaking at the qualifier does not. Panel B shows the twelve highest-volume Boston-anchored course pairs in the full international estimation panel.

generates little displacement, and runner-clustered bootstrap intervals are tight. Finally, the estimand is a course as historically run: route changes and young courses motivate regular re-estimation rather than interpreting each effect as an immutable property of the terrain.

The third assumption is that movement is unrelated to the transitory part of performance. Persistent sorting of strong runners into fast courses is not the problem: runner fixed effects absorb persistent ability. The problem is differential temporary state. Runners choose qualifying venues for the attempts when they are trained, tapered, healthy, and close to their standard. A naive course effect can then combine the venue's systematic advantage with a qualifying campaign.

This is the Rothstein critique of value-added measurement (Rothstein, 2010; Chetty, Friedman and Rockoff, 2014): an estimated environment effect can reflect selection on the transitory component of performance rather than the environment itself. The incentive is a public threshold that moves on announced dates; venues are discrete and physically measurable; weather is observed; and the same runner appears on both sides of a switch. Contamination should therefore be directional, concentrated near the threshold, and strongest at qualifying venues. The data show all three: runners under-transfer from qualifiers into Boston, while reverse moves and the lottery-entry city majors (Chicago, New York) do not show the same pattern; Marine Corps and Philadelphia, goal races for part of their fields, sit near Boston's slope (Appendix Table B3). The violation is systematic enough to estimate and correct: the repeat-runner design remains a valid measurement device once the venue-specific share of strategic attempts is used to scale the correction.

BQ-induced contamination can enter a course effect only through runners who use that course strategically. Let s_c denote the share of course c's finishers whose qualification depends on the course. To first order,

$$\hat{\delta}_c^{\text{naive}} \;=\; \delta_c \;+\; s_c\, b_c \;+\; \text{noise}, \tag{2}$$

where $\hat{\delta}_c^{\text{naive}}$ is the estimated, uncorrected course effect of equation (1), δ_c is the systematic course advantage net of strategic effort—terrain, persistent climate, and race operations—and b_c is the average behavioral residual of strategic users. The median course with a measured peaking residual has a strategic share of about 0.9%. Contamination is therefore concentrated at the dedicated qualifying products rather than spread uniformly across the course index. Directional mover pairs estimate b_c, and observed strategic shares supply s_c. Routine attempts and reverse pairs check each independently.

For an ordered same-runner pair $c \to c'$, define the residual $\eta_{c\to c'} = (\ln T_{c'} - \ln T_c) - (\hat{\delta}_{c'} - \hat{\delta}_c)$. Its mean is near zero, but it is systematically positive into Boston: runners are +7.9 minutes slower there than the additive model predicts, versus -2.1 minutes when Boston is the origin, an asymmetry of +10.1 minutes (Figure 2, Panel B, and Appendix Table C1). The asymmetry is absent for lottery-entry destination races and is largest when the runner qualified at the origin, exactly as targeted effort predicts. Directional mover residuals provide the course-level estimate of that behavioral component.

At St. George, a steeply downhill course, course-dependent qualifiers are about 5.8% of finishers; the median among courses with a measured peaking residual is 0.9%; only the dedicated qualifying products reach contamination-relevant shares (the fastest courses' shares are tabulated in Appendix Table C2). Because strategic users are a minority almost everywhere, they barely move the course effect: re-estimating the model after excluding near-threshold observations (the "clean" index) leaves the benchmark downhill courses essentially unchanged. A steeply downhill course is genuinely fast for everyone who runs it.

The behavior-adjusted index implements equation (2), netting out the course-level peaking contamination by scaling the origin peaking residual $\eta_c \equiv \eta_{c\to\text{Boston}}$ by the strategic share:

$$\hat{\delta}_c^{\text{adj}} = \hat{\delta}_c^{\text{naive}} + s_c\,\eta_c. \tag{3}$$

The into-Boston residual η_c estimates the negative of b_c in equation (2) (a course whose users peaked there looks too fast, and η_c is positive), so adding $s_c\,\eta_c$ undoes the contamination term $s_c\,b_c$. The correction is estimated course by course and is not one-signed. At the last-chance products with the highest strategic shares, the estimated peaking residual is negative: runners who qualified there went on to run Boston somewhat faster than the additive model predicts, so the adjustment credits these courses with more speed than the naive index reports (Table C2). The behavioral correction therefore does not mechanically shrink the qualifying products' advantage; where the Boston retest validates the transferred advantage, the adjusted index moves the other way. The adjustment is deliberately modest and agrees with the clean re-estimate: St. George moves from −6.4 to −5.8 minutes relative to Boston, and the typical course barely moves. Only a course's strategic users additionally peak, so only their share of the field is netted out. Applying the raw retest residual to the whole course instead would be a selection error: the residual is estimated from the self-selected minority who qualified and went to Boston, and attributing their behavior to every finisher would flip genuinely fast courses to slower-than-Boston. Section 2 uses the behavior-adjusted index as the headline screen, with the full suite as the robustness band.

The correction is monotone in the strategic share, exceeds one minute only at high-strategic- share courses, and is below one minute at courses accounting for 99% of U.S. road finishes (Appendix Table C2). Thus the naive and behavior-adjusted indices are nearly identical for ordinary venues; the meaningful exceptions are the qualifying products themselves.

Across all 7 indices, built on one consistent sample, displacement ranges from 9.0% to 12.5%; the policy accounting of Section 2 reports its headline against this band. Appendix Table C3 collects the full suite: each index's construction, the bias it addresses, its course coverage, its rank correlation with the headline index, and the fixed-field displacement it implies. Two members of the suite are directly implementable as rules: the conservative shrinkage and the observable-predicted formula, which needs no race history and so covers brand-new courses; Appendix C details both and the practical hybrid that combines them. The band has one limitation: the observable-predicted index is fit to contaminated fixed effects, and peaking correlates with fast terrain, so it is not a fully behavior-free benchmark either.

2 Who Qualifies Because of the Course?

By the most recent trusted cohort, nearly three in ten linked Boston entrants with an observed effective qualifying performance have no observed performance that would still qualify after Boston-equivalent adjustment. I measure the distortion's scale and growth, decompose it into the formal rule, venue supply, and runner sorting, and construct the counterfactual field under a venue-adjusted standard; participation selection is examined separately in Appendix E. Two statistics recur and answer different questions. The course-dependence rate just quoted counts entrants whose qualification depends on their venue at the current cutoff, among the linked entrants who can be classified. The fixed-field displacement rate counts the qualifications a venue-adjusted ranking would reallocate; it is smaller both because it is computed on the full qualifying field, where fast-course qualifiers are less over-represented than among realized entrants, and because holding the number of qualifiers fixed loosens the venue-adjusted cutoff, since fewer runners from slower courses clear the current bar after adjustment than fast-course runners fall below it. The benchmark is the principal's own stated objective—select the fastest runners on a common basis ("earn a spot")—so misallocation here means deviation from that meritocratic objective, not a social-welfare claim; the BAA may also value revenue or geographic breadth, which would change the normative reading but not the positive results. All accounting in this section runs on the US road-course universe.

For each Boston entrant, I first ask whether any observed qualifying-window performance clears the effective cutoff; if so, the entrant enters the classifiable sample. I classify the entrant as course-dependent when none of the entrant's observed performances would clear the effective cutoff on a Boston-equivalent course. This uses every observed US-road performance in the window, not only the fastest or any selectively chosen race, and counts each entrant once. BQ age is age on Boston race day. Entrants with an unindexed observed performance are reported as unclassified. A separate performance-level series uses the posted standard to describe the larger population responding to the public target; it should not be read as the realized Boston field.

Figure 3 plots the posted-standard performance series alongside the direct entrant-level effective-cutoff series. The former rises from about 9.4% in the Boston 2004 cohort to 17.5% in 2019 and 17.4% in 2025. Among linked, classifiable Boston entrants, the effective-cutoff rate rises from 14.5% in 2004 to 28.5% in 2019 and 29.1% in 2025, the most recent trusted endpoint; essentially all of the growth arrives by 2019, and apart from a dip in the COVID-disrupted 2022 cohort the rate has since held near its elevated level. The classifiable sample covers 49.5%, 44.6%, and 40.0% of the observed Boston fields in 2004, 2019, and 2025; unmatched entrants remain unclassified rather than being coded as unaffected. Boston 2026

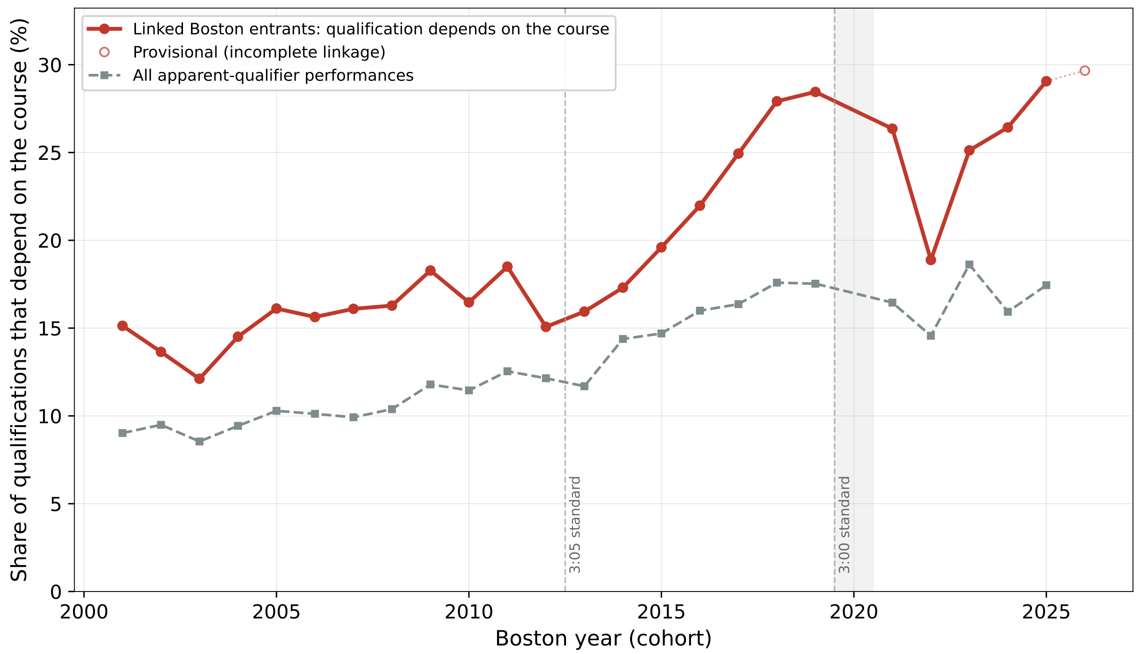

Figure 3: Course-dependent qualification has risen. The share of each population whose qualification depends on its course: linked, classifiable Boston entrants at the effective admission cutoff (solid) and apparent qualifying performances at the posted standard (dashed). The shaded band marks the cancelled 2020 race; Boston 2025 is the latest trusted endpoint and 2026 is shown as provisional (open marker).

is retained at 29.7%, but is marked as provisional because its external course-level linkage diagnostic is less uniform; its classifiable sample covers 37.2% of the full field.

Re-classifying every entrant with the behavior-adjusted index moves the series only slightly, from 13.5% to 27.3%. The observable-predicted index, which attributes to the venue only what physical features predict, puts the level lower (8.2% to 15.8%) but preserves the doubling. And within the linkage-homogeneous US-resident subpopulation, where match rates are uniform across cohorts, the series is if anything slightly steeper (14.6% to 30.8%), so the decline in classifiable coverage is not manufacturing the trend. The level depends on how much of a course's advantage the index attributes to the venue; the doubling does not. The same shift appears in course terrain: the share of the Boston field qualifying on any net-downhill course rose from about 30 to 42%, and on courses dropping more steeply than Boston itself from 9 to 21%.

Venue-adjusted standards address the screening distortion. By removing the qualifying advantage of fast venues, they would also weaken runner sorting and venue entry over time. The exercises here are partial equilibrium—course choices held fixed—so they measure the immediate reallocation, with an equilibrium-response bound at the end. They reallocate qualification, not Boston entry: who actually starts depends additionally on take-up, which differs by course and would itself respond to the rule.

The natural rule adjusts each course's qualifying times by its estimated speed: proportional handicaps, applied uniformly across age and gender groups. Most courses are in fact slower than Boston, and their runners are penalized by the common standard just as fast-course runners are subsidized; a venue-adjusted standard corrects both directions, while discussion of the downhill extreme misses the other half of the reallocation.

Holding the number of qualifiers fixed and ranking all candidates by behavior-adjusted margin (the index constructed in Section 1; the full suite of corrections bounds the result), 10,354 runners—11.9% of the 87,066-runner 2024–25 qualifying field (bootstrap 95% CI 11.7–12.0%)—would be displaced and replaced one-for-one. The displaced come overwhelmingly from fast courses (88.3%); the replacements come from courses at Boston's pace or slower (Appendix Figure F1). Displacement is 9.0–12.5% across the full index suite (Appendix Tables C3 and F1 report every index and rule), so the conclusion survives every correction of Section 1.

Two further checks address estimation error, because ranking by $T_i e^{-\hat{\delta}_{c(i)}}$ converts noise in $\hat{\delta}$ into mechanical displacement. First, a leave-out design: re-estimating the index with 2024–25 excluded (so that no qualifying performance being ranked contributes to the course effects that rank it) leaves displacement essentially unchanged, 12.4% versus 12.4% on the identical field (course-level correlation 0.982 between the two indices). Second, a pure-noise placebo: simulating course effects $\delta_c \sim N(0, \widehat{se}_c^{\,2})$ under zero true venue heterogeneity generates displacement of only 1.4% (95% range 1.1–1.8%), an order of magnitude below the headline. Nor is displacement clustered at the cutoff: the median displaced runner was 2.9

minutes inside the posted standard, only 20% were within one minute, and replacements sat a median 4.0 minutes outside it, with course-neutralized times 6.1 minutes faster than the runners they displace.

The venue-adjusted qualifier pool is also predicted to be faster at Boston. Using each runner's permanent ability (runner effect plus age–gender profile) to predict Boston-day performance, replacement qualifiers are predicted to run about 4.9 minutes faster at Boston than the runners they displace (Appendix Figure D3): the displaced qualify on venue, not on ability.

3 Runners Respond—and Races Enter

Runners and race organizers respond to the qualifying incentive; this section documents that behavior, on the same US road-course universe as Section 2. Throughout, the design contrasts populations that face the qualifying incentive with populations that do not: near-threshold versus inframarginal runners, BQ thresholds versus round-number placebos, announced tightenings versus surrounding years. Every estimate uses only the coarse fast/slow course classification, which is robust to all of the index corrections in Section 1; because the estimates measure deliberately different estimands on different populations, Appendix Table E3 maps each to its object.

Runners near the qualifying threshold concentrate on faster courses, and the concentration has grown (Figure 4, Panel A). The share is flat far from the threshold and rises sharply near it; the near-threshold excess grew era by era: 4.6, 7.9, and 10.9 percentage points relative to each era's field in the 3:10, 3:05, and 3:00 eras. Inframarginal runners, too far from the cutoff for course choice to matter, show no comparable gradient, which addresses amenity- and geography-based alternatives: both groups face the same courses, travel costs, and calendars; only one faces the incentive.

Treating the inframarginal venue distribution as the no-incentive counterfactual, roughly

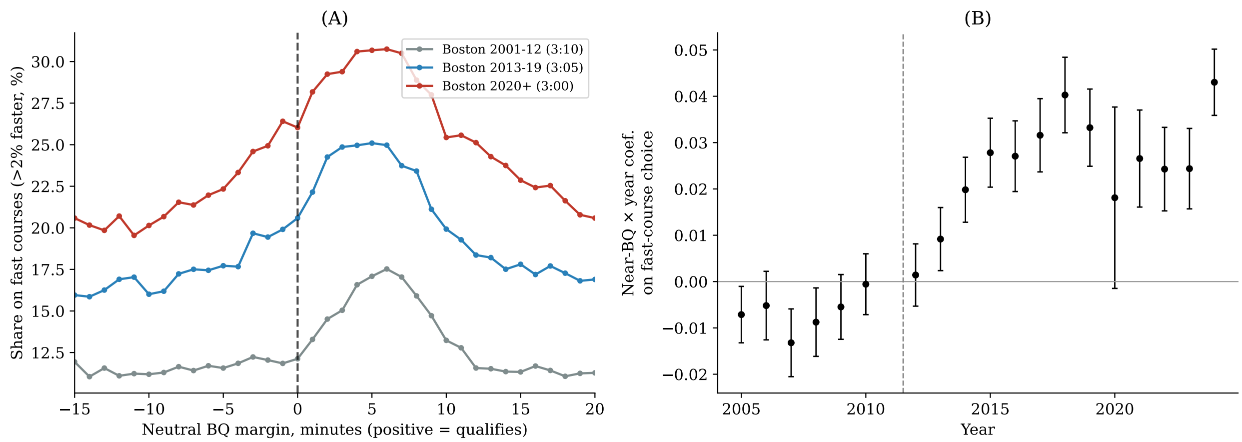

Figure 4: Runners close to the Boston standard increasingly choose fast courses. Panel A reports the fraction choosing a fast course by Boston-equivalent minutes from the runner's own standard, by qualifying regime; sorting is concentrated around the economic margin and the near-threshold gradient grows across eras. Panel B is the event study of near-BQ runners' differential fast-course choice, relative to 2011 (95% CIs, runner-clustered): no comparable pre-announcement increase, and a sustained rise after the February 2011 announcement (the first post-announcement full qualifying year is 2012).

one in five near-BQ runners is at a course they would not otherwise have chosen (Appendix Figure E1). Take-up is consistently higher at faster courses among runners who clear the effective cutoff, consistent with revealed BQ intent (Appendix E reports the take-up accounting).

Runners whose previous performances put them under greater BQ pressure subsequently choose measurably faster venues, and the fast-course share peaks for runners whose prior best just missed the standard (Appendix Table E1). The test parallels the forecast-bias test of Chetty, Friedman and Rockoff (2014), with predictable selection as the outcome. A conditional-logit model of venue choice, reported in the replication archive, quantifies the same pressure response.

In an event study of near-BQ runners' differential fast-course choice by year, the differential is flat before the February 2011 announcement and rises steadily afterward (Figure 4, Panel B); the compact difference-in-differences coefficient is +3.1 percentage points (SE 0.16; Appendix Table E1) and survives home-state and year fixed effects. Because a fast course lies within 150 miles of 77% of US-resident runners, geographic access does not constrain the response.

Marathon finishers bunch at salient round numbers (Allen et al., 2017). Boston-qualifying times are not round for most demographics, providing a test of whether the threshold itself is a target. Three facts show that it is, and that the bunching is concentrated at fast venues. First, the bunching spike tracks the standard as the BAA moves it: re-centering every demographic's finishes on its own pre-2011 standard, the fast-course density kinks exactly at the active BQ in every era, five minutes further left after each tightening, while the slow-course density stays smooth (Figure 5). Because the spike moves with the cutoff, it cannot simply reflect a fixed feature of these courses or runners. Second, regressing each marathon's estimated BQ-bunching premium on its course speed, the slope is -0.0131 per minute of course speed after the 2011 announcement (permutation $p < 0.001$, reassigning course speeds across marathons) versus -0.0088 before: the course-speed gradient in BQ targeting is a post-tightening phenomenon. Third, excess-mass estimates—the standard bunching estimator, which measures the extra density piled against the threshold relative to a smooth counterfactual (Saez, 2010; Kleven and Waseem, 2013)—by course type and era (Appendix Table E2) show that before 2012 the fast$-$slow excess-mass gradient at the BQ threshold is roughly four times the corresponding gradient at the 4:00 round-number placebo, and that after 2012 the placebo gradient disappears entirely while the BQ-threshold gradient persists (Appendix Figure E2); the cell-level interaction versions of these contrasts are signed identically but individually imprecise, and inference rests on the marathon-level gradient above.

Fast courses expanded 1.9-fold, and purpose-built qualifying products entered on a timeline aligned with announced tightenings.

The number of US courses at least five minutes faster than Boston at the M18–34 standard (about 3%, a stricter bar than the 2% "fast" classification used elsewhere in the paper) rose from 23 in 2004 to 44 in 2025, with the steepest growth between the 2011 and 2018 announcements (Figure 6). The fast-course share of US finishers rose from 5.7% to 11.1% over

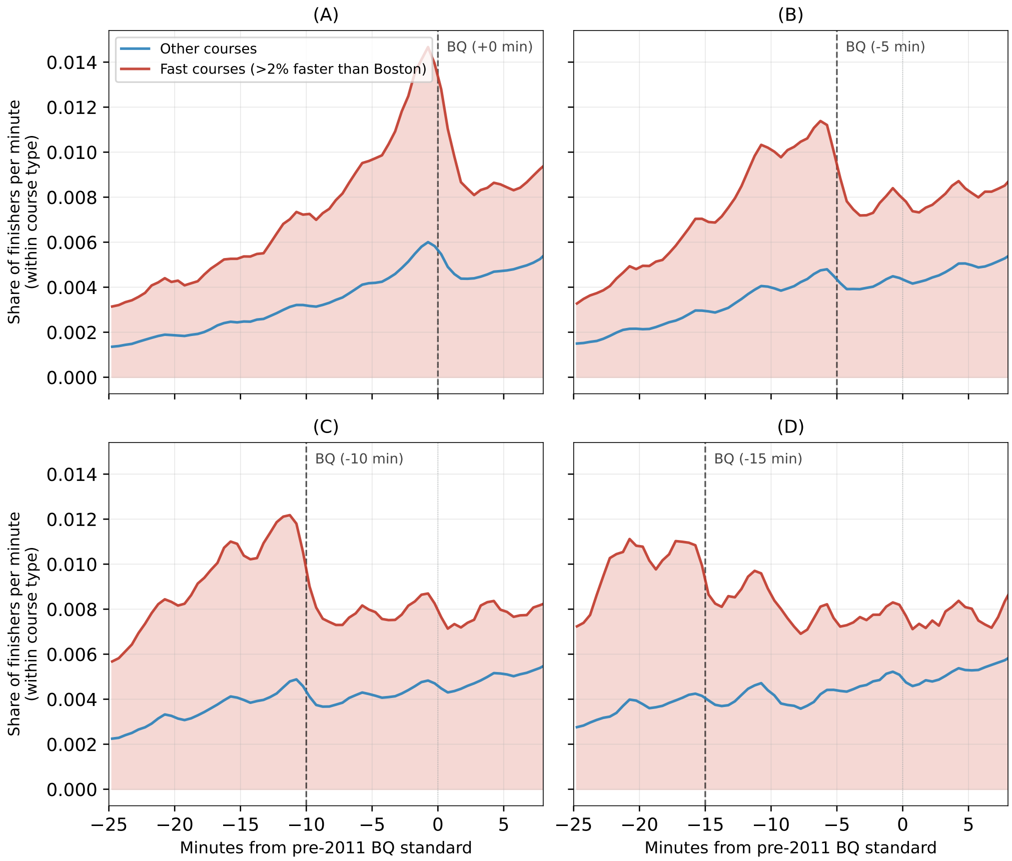

Figure 5: BQ bunching is concentrated at fast courses. Each panel shows one qualifying era, with finish times re-centered on the demographic's pre-2011 standard, so the successive standards appear at offsets of 0, −5, −10, and −15 minutes. In every era the fast-course density spikes at the standard then in force and the spike moves with each five-minute tightening, while the density at other courses remains comparatively smooth.

the same period. Entry concentrates in the product category the rule rewards: the REVEL series (purpose-built point-to-point downhill races, multiple launches 2014–2018) and the "BQ.2" races (Spring Chance, Last Chance Chicagoland, Last Chance Grand Rapids), whose branding ("BQ" is the product name) markets the screening rule itself.[3]

Appendix Figure E3 shows the timing and composition of entry. The fast share of newly entering US courses rises by roughly half, from about 18% of entrants before 2008 to about 26% afterward (peaking at 29% in 2011–2018), with the very-fast share more than doubling, from 5.5% to 14.2%. The flagship launches land almost entirely between the two

[3]The series in Figure 6 uses the five-minute definition; the 2% classification used elsewhere in the paper shows the same proportional growth.

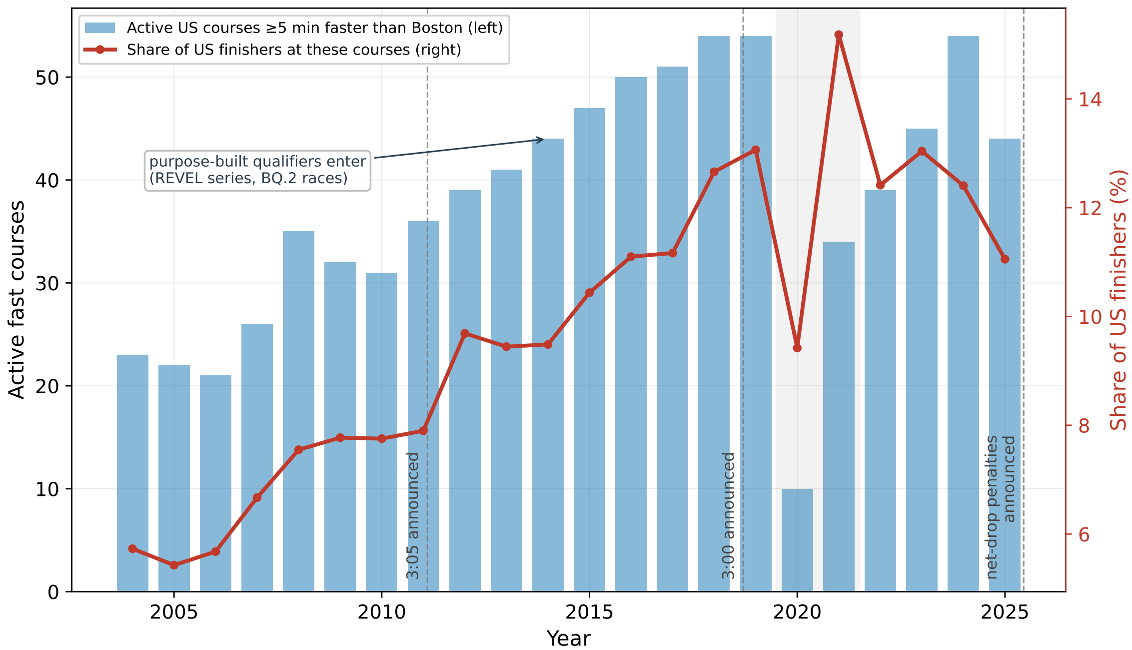

Figure 6: Entry and growth of fast venues. Bars count active US courses at least five minutes faster than Boston at the M18–34 standard; the line is the fast-course share of US finishers. Dashed lines mark the February 2011 and September 2018 announcements and the June 2025 net-drop rule; the shaded band marks the COVID-disrupted 2020–21 races.

announcements: nine purpose-built products enter from Big Cottonwood (2012) to Spring Chance BQ.2 (2018); the figure names each. (The post-September-2018 event window falls in the COVID years, so the 2011 announcement provides the clean timing test.) Entry selection toward fast products begins around 2008, after Boston's registration sellouts made the demand pressure visible and before the announcement that codified it, consistent with entry responding to the incentive itself rather than its public signal.

In the Boston 2025 cohort, 43% of course-dependent qualifiers (those who clear the standard only by virtue of their course) ran a course that did not exist in 2010, versus 18% of all apparent qualifiers. Entry and runner sorting respond to the same underlying BQ incentive, so the decomposition below is accounting rather than causal identification.

Venues created to sell qualification convert qualifiers into Boston entrants at 1.9 to 2.6 times the take-up rate of the majors, consistent with runners choosing these venues specifically to enter Boston.

To separate the pre-participation channels, I build a chain of counterfactuals. Start with the runners in each Boston cohort, place them at random into the frozen 2005–07 course market, and apply the 2005 effective cutoff. Then tighten the cutoff to the cohort's own (step 2). Then let the course menu expand to the races that had actually entered by that year (step 3). Finally, let runners choose the races they actually chose (step 4). The linked-entrant series is not an additional additive channel: it replaces performance-level accounting with the direct entrant-level measure above, counting linked Boston entrants once. The step-1-to-step-4 differences are path-dependent accounting contributions of the formal rule, venue supply, and runner sorting, with a common upstream driver; participation selection is examined separately in Appendix E by comparing take-up among actual effective qualifiers. Because the baseline is random assignment itself, the sorting residual is the total deviation from random, not just its growth since the base period. The chain uses observed course-adjusted times (not estimated runner effects), so noise in the runner fixed effects does not enter the simulated qualification rates; noise in $\hat{\delta}$ enters through neutralization and is bounded by the index band of Section 1.

Table 1 reports focal years; Appendix Figure D4 plots the full chain by cohort. Under random assignment to the frozen menu, misallocation is modest and roughly flat. The direct rule effect contributes about 3.1 percentage points by the 2025 cohort. The venue-supply channel—applying the same random assignment to the actual, expanding course menu—contributes 6.2 points under this baseline; runner sorting beyond random assignment adds 6.0 points. Under the inframarginal baseline used for the headline numbers in the introduction, the corresponding contributions are 5.5 and 7.2 points. The two endogenous market channels together dominate the formal threshold change.

Table 1: Misallocation accounting at focal years. Steps 1–4 form the random-assignment chain. The separately labeled linked-entrant column reports the direct Boston-field estimand; it is not an additional additive channel.

	Rates (%)					
Year	s_1 Random	s_2 +Threshold	s_3 +Menu	s_4 +Sorting	Linked entrants	Coverage
2004	6.2	6.2	6.1	9.4	14.5	49.5
2010	6.2	6.2	7.1	11.5	16.5	51.1
2015	5.8	7.1	10.3	14.9	19.6	50.6
2019	5.3	7.5	12.5	20.2	28.5	44.6
2024	3.9	6.5	11.3	15.7	26.4	35.9
2025	4.1	7.2	13.4	19.4	29.1	40.0
2026^p	4.2	7.9	13.3	19.4	29.7	37.2

Notes: Misallocation rates in percent, US road-course qualifying universe, course effects measured relative to Boston. Step 1 assigns each cohort's runners randomly to the frozen 2005–07 course menu at the 2005 effective cutoff; steps 2–4 add the cohort's actual cutoff, menu, and assignment. The linked-entrant column counts Boston entrants with an observed effective qualifier once and requires no Boston-equivalent effective qualifier anywhere in the observed window. Coverage is the classifiable share of the observed Boston field. ^{p} The 2026 cohort is provisional: its course-level linkage diagnostic is less uniform than earlier cohorts.

The random-assignment baseline is deliberately neutral, and it is not innocuous: people run their local race, so part of the step-4 deviation from random assignment is geography rather than incentive. Two geography-respecting alternatives (assigning each cohort's runners to the venue distribution of its inframarginal runners, and a gravity baseline with distance decay calibrated to observed travel) raise the sorting channel rather than explain it away, because full-field venue shares already contain the marginal runners' own sorting. Across every baseline and ordering of the chain, sorting contributes 6.0–9.6 points and supply 3.0–6.2: sorting is at least as large as supply in all but one baseline–ordering combination, and venue supply is never negligible (Appendix Table D1 reports the full grid).

4 Boston Changed the Rules

Boston has already changed the rules. Beginning with the 2027 race, the BAA will add five minutes to qualifying times from courses with 1,500–2,999 feet of net descent, ten minutes for 3,000–5,999 feet, and will not accept times from courses dropping 6,000 feet or more. The policy change formally recognizes that qualifying times from different courses are not directly comparable. The question is how much of the venue advantage the announced rule actually corrects. I compare four choices on one fixed field: retain a common cutoff, apply the BAA's one-sided net-drop penalties, use a simple symmetric physical formula, or handicap qualifying times with the behavior-adjusted course effects. The comparison is conducted on U.S. road marathons, the relevant qualifying market. Trail and mountain races remain useful for understanding measurement but should not determine the road-marathon policy.

I use the behavior-adjusted course-effect handicap as the benchmark for comparability: adjusting every submitted time by the estimated systematic course advantage ranks applicants as though they had competed under common conditions, and I call the resulting qualifier set the common-course benchmark. The adjustment is symmetric: fast courses receive penalties, slow courses receive credits, and runner effort remains part of performance. Most major city marathons are slower than Boston, so under consistent venue pricing their qualifiers would receive credits, not penalties. The benchmark also embeds a normative choice, since the course effect bundles terrain with persistent climate and race operations, and the BAA could reasonably decide to neutralize only some of those components; Section 1 already strips the behavioral component, which no reasonable rule should penalize.

Turnover relative to the status quo is not itself a measure of accuracy: a rule that changes few runners could change the right ones, and a rule that changes many could change the wrong ones. The relevant measure is how far each rule's qualifier field remains from the benchmark field. Table 2 reports both. With no adjustment at all, 11.9% of the fixed field differs from the common-course benchmark. The BAA's announced penalties lower that disagreement to 11.1%, closing 6.3% of the gap (4.8–7.6% across leave-out and corrected-index benchmarks). The rule is well targeted but far too narrow: 87% of the performances it rejects are ones the benchmark also rejects, but the penalized courses account for only 3.1% of qualifying-market finishes, and just 53% of the replacement qualifiers it promotes are the benchmark's replacements. The symmetric elevation formula closes 22.5% of the gap. Any change also needs a transition rule: the displaced would be runners who followed published rules in good faith, so a new standard would sensibly apply prospectively, to qualifying windows announced in advance, as the BAA's own 2027 rule does.

Table 2: How close each rule comes to the common-course benchmark. The fixed field holds the number of qualifiers at its actual level; each rule re-ranks the same performances.

Rule	Field changed (%)	Disagreement with benchmark (%)	Gap closed (%)	Repl.–displ. gap (min)	Symmetric
Common cutoff	0.0	11.9	0.0	–	No
BAA 2027 net-drop	1.9	11.1	6.3	8.7	No
Symmetric elevation	6.7	9.2	22.5	6.2	Yes
Course-effect benchmark	11.9	0.0	100.0	4.9	Yes

Notes: US road qualifying market, 2024–25 performances, field size 87,066. "Field changed" is each rule's one-sided turnover relative to the common cutoff. "Disagreement with benchmark" is the share of the field that differs from the common-course benchmark's qualifier set, $|S_r \triangle S^*|/2N$. "Gap closed" is the share of the common-cutoff disagreement a rule eliminates, $1-$ disagreement/11.9%. "Symmetric" indicates whether the rule both penalizes fast courses and credits slow ones. The replacement–displaced gap is the difference in AKM-predicted Boston-day margins between the performances a rule adds and those it removes (positive values mean the added performances are predicted faster at Boston); it is computed over each rule's own swaps, so it mechanically shrinks as a rule reaches deeper into the field and is not a ranking criterion. The benchmark row is exact by construction; scoring the rules against a leave-out re-estimate of the course effects, or against any index in the corrected suite, changes the BAA rule's captured share by at most a few percentage points (4.8–7.6%).

The BAA's 2027 rule penalizes net elevation drop alone. It has practical advantages: it is simple, it can be announced two years in advance, and it directly deters the extreme-downhill product category. As a measurement device, however, it is weak. Among profiled

U.S. road courses, net drop explains only 0.07 of course-speed variation out of sample, compared with 0.33 for gross climbing and 0.47 for the full physical-and-weather formula. Even that formula explains only 0.12 of the variation weighted by where runners actually race. A one-sided net-drop rule therefore misses on both sides: fast-flat qualifying products escape unpenalized, while some large-net-drop courses with heavy climbing are overcorrected. REVEL Mt Charleston drops 5,200 feet and draws the 10-minute penalty, but its estimated advantage at the qualifying margin is about 17 minutes, so about 7 minutes of the advantage survive. The Spring Chance BQ.2 is essentially flat, with an advantage of about 17 minutes, and receives no penalty at all. Big Cottonwood is penalized 5 minutes against an advantage of about 9. And New York City, about 4 minutes slower than Boston, receives no credit. Of the 75 fast US courses active in 2024–25, 63 (about 84%) fall below the 1,500-foot net-drop floor and are untouched by the rule. The framework predicts that entry, course design, and marketing will pivot toward this unpenalized tier. Suppliers are already responding: one steep Utah course markets its 2026 edition as the last fast qualifier before the penalty takes effect. Any narrow proxy also creates a new target, inviting re-sorting toward whatever favorable venues the formula leaves unpenalized.

Keeping the common cutoff is maximally transparent but preserves the distortion. A cardinal handicap based on this paper's Empirical-Bayes course index best equalizes measured venue speed, but inherits estimation error and must be re-estimated as courses change. A symmetric physical formula using gross climbing and descent, maximum elevation, and typical weather is easier to publish and can score new courses, but its out-of-sample fit is materially weaker. The BAA's one-sided net-drop proxy is simplest and least accurate. Corrections must also choose which components to neutralize: the systematic course advantage but not effort.

Reweighting near-BQ runners to the inframarginal course distribution (the no-incentive counterfactual) removes about half of the near-BQ fast-course excess and lowers the misallocation rate among apparent qualifiers by roughly five percentage points in 2018–19.

Equilibrium exit of marginal demand would additionally discourage fast-course entry, so the fixed-choice numbers above are a lower bound on the policy's full effect. The bound's assumption is that inframarginal runners' venue distribution is not itself fast-tilted for non-BQ reasons. Finally, any published handicap itself becomes a target: runners can chase courses with imprecise or stale handicaps, and suppliers can design to the formula. How often the index is re-estimated, whether handicaps apply ex ante or ex post, and how new courses are scored will determine how much of the screening distortion survives. A static handicap reduces today's misallocation; it does not prevent runners and suppliers from adapting to it.

5 Lessons for High-Stakes Measurement

A common qualifying threshold applied across heterogeneous, manipulable evaluation environments produces two linked distortions. The venue value-added a naive fixed-effects index reports is a composite of systematic venue advantage and the behavior the threshold induces.

The screening distortion is large. Among linked Boston entrants with an observed effective qualifier, 29.1% in the trusted 2025 cohort have no observed performance that would still clear the effective cutoff on a Boston-equivalent course, roughly double the 2004 share, and adjusting qualifying times for course speed would change about 11.9% of who qualifies, with the newly qualifying predicted to be faster at Boston itself. The market distortion is two-sided: near-threshold runners sort to fast venues, with the response growing after announced tightenings, and purpose-built qualifying products entered until the supply channel contributed misallocation of the same order as candidate sorting.

These distortions cannot be repaired by targeting a few extreme courses. In the policy-relevant U.S.-road sample, gross climbing and typical weather predict speed better than net drop, but even the full physical formula leaves most finisher-weighted variation unexplained out of sample. Index-based venue adjustment is therefore the most accurate rule for established courses, with a symmetric physical formula as a transparent fallback for new ones.

In most settings where a measure becomes a target (Goodhart, 1984; Campbell, 1979), only part of the response is observable. Here all of it is: the rule, the runner and organizer responses, the physical determinants of venue speed, and a retest at Boston itself. When qualification depends on an estimated fixed effect, runners move toward favorable venues, concentrate near the threshold, and respond to each tightening. Those regularities are what make the problem correctable: distance-to-threshold comparisons and moving-threshold placebos detect the behavior, directional mover asymmetries estimate it, and strategic shares scale the correction. Audits of candidate gaming alone miss the supply side, which here grew to the same order as candidate sorting within a decade.

Detection is easier here than in most applications: the threshold is public, sharp, and moves on announced dates. Where the operative threshold is vaguer—teacher evaluation, hospital report cards—the same estimates will be noisier. Three strategies do not depend on the marathon setting: measuring the physical drivers of environment advantage directly, using the institution's own announced rule changes as the source of variation, and examining supplier entry alongside candidate behavior. In any such setting, a common threshold rewards the environment as well as the person, and candidates and suppliers will find the environments it rewards.

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Appendix roadmap

Main-text claim	Supporting appendix material
The data, samples, and runner linkage are sound	Appendix A
Course effects identify persistent course advantage	Appendix B
Strategic effort does not drive the course rankings	Appendix C
Course dependence and field reallocation are large	Appendix D
Runners sort and venues enter	Appendix E
The BAA rule captures little of the benchmark correction	Appendix F

Appendix A Data Sources and Sample Construction

The introduction describes the primary data sources and estimation sample. This appendix documents the sources, the construction of the samples used later in the paper, the entity-resolution procedure, and the selection of the race-time variable.

A.1 Sources

Table A1 lists each data provider: its access method, the years it covers, and the marathons, race-years, and observations it contributes to the combined panel. Each (marathon, year) cell enters from exactly one source; where sources overlap, a fixed priority ranking prefers race-specific and organizer sources over aggregators.

Table A1: Data sources and their contribution to the combined panel

Provider	Access method	Years	Marathons	Race-years	Observations
Bulk US results archive	Aggregated bulk file	2004–2025	819	5,935	6,299,304
Athlinks (athlinks.com)	Results API	2000–2026	557	1,779	3,250,386
MarathonGuide (marathonguide.com)	Results API	2000–2025	633	1,672	2,550,808
Results.com bulk file	Bulk file	2010–2019	89	501	1,595,049
mikatiming (Berlin, Stockholm, Hamburg, Frankfurt, Boston live)	HTML results pages	2004–2025	5	37	869,047
Boston Athletic Association results service	Results API	2001–2026	1	25	564,359
Race-organizer files (Chicago, Marine Corps)	Files from organizers	2000–2015	2	28	762,472
Official race results services (Paris, Tokyo, Prague, REVEL, Munich, Zurich, Cape Town, others)	Race-specific scrapers	2000–2026	21	137	1,313,237
Hand-collected small races	Posted results files	2015–2024	2	15	7,802
Total		2000–2026	1,172	10,129	17,212,464

Notes: Counts are rows surviving the one-source-per-(marathon, year) deduplication, so observations sum to the panel total. Two collected sources (legacy Boston year files and Manchester 2000–02) are fully displaced by higher-priority sources and contribute no surviving rows. All sources provide publicly posted results with at least name, gender, age or age group, and finish time. Beyond those common fields, Athlinks and MarathonGuide add placings, chip and gun times, and residence; mikatiming and the race-organizer files add year of birth; and the BAA service adds race-day age, residence, citizenship, and split times. Individual finish records are not redistributed; the replication archive contains the code and derived course-level datasets.

A.2 Sample Construction

Table A2 summarizes the construction of the samples used later in the paper, and the sample sizes differ across rows for the following reasons. The raw linked panel is the full web-scraped dataset after entity resolution: every finish-time observation at every marathon with runner identities assigned across races. The AKM estimation sample is smaller because the model requires each runner to appear at least twice within a geographic region so that the runner fixed effect is identified, requires a known year of birth so that the age polynomial is defined, and trims the slowest 0.1% within each race-year to prevent outliers from dominating the residual variance. These three filters drop singletons, observations with missing year of birth, and extreme outliers, which together cut the observation count from 17.2 million to 11.1 million. The course count falls modestly because small races at which every runner is a singleton contribute no observations after the two-appearance requirement. The course speed index row is a course-level object, not an observation-level one; it reports only the number of courses retained for the published index, which are those with at least 50 AKM observations after shrinkage. The qualification analysis sample is the set of non-Boston observations at indexed courses in years with a known BQ standard. It is drawn from the full linked panel rather than the AKM sample, so singletons reenter (their BQ margin is defined without a runner fixed effect) and the observation and runner counts rise correspondingly. The cross-race additive separability pairs are a pair-level object: for each runner-race in the linked panel, I form a pair with every prior marathon by the same runner at a different course within 18 months. The runner count is lower than in the AKM sample because only runners with at least two appearances within an 18-month window can generate a pair. The Boston-destination pairs are the subset whose destination is Boston, which defines the standardized retest because the index is normalized so that $\hat{\delta}_{\text{Boston}} = 0$.

Table A2: Sample construction

Sample	Observations	Runners	Courses
Raw linked panel	17,206,046	8,124,265	1,172
AKM estimation sample	11,142,687	2,631,055	1,162
Course speed index	—	—	1057
Qualification analysis	9,410,636	3,949,983	721
Cross-race additive separability pairs	10,934,525	—	—
Boston-destination pairs	690,058	—	—

Notes: The raw linked panel is the web-scraped dataset after entity resolution. The AKM sample requires at least two appearances per regional runner, a known year of birth, and trims the slowest 0.1% within each race-year. The course speed index retains courses with at least 50 AKM observations after Empirical Bayes shrinkage. The qualification analysis uses all non-Boston U.S. road observations at indexed courses with BQ standards; its course count is smaller than the index because non-US and trail courses and courses without indexed years in the qualification window drop out. Additive-separability pairs link each runner's destination race to every prior marathon at a different course within 18 months; Boston-destination pairs restrict the destination to Boston. Samples are used as follows: the raw panel in Sections 2 and 3; the AKM sample in Sections 1 and 4; the qualification sample in Sections 3–4; the pair samples in the validation and ordered-pair diagnostics of Section 1; and the geocoded US subpopulation in Section 3 (selection and choice analyses).

A.3 Entity Resolution

I link runner records in five steps, beginning with a high-recall merge and then pruning implausible groupings using rarity and within-group contradictions:

1. Rarity tables and race communities. Build name-rarity tables (counts of observations sharing a given canonical first name, last name, gender, and year-of-birth bucket) so that downstream decisions can condition on how distinctive a given identifier combina-

tion is, and construct a graph of marathons that share unusually many high-confidence linked runners; connected components of this graph define "communities" of races that recruit from overlapping runner populations.

2. High-recall merge. Form initial runner clusters by merging observations that share a canonical first name, last name, and gender, with year-of-birth resolved at the runner level: missing years of birth are imputed from age bands or from peer observations of the same canonical identity, and the merge requires that all observations sharing an identity have year-of-birth ranges that intersect. Country and state are imputed where partially observed, and runner-level country is propagated within the cluster.

3. Contradiction splits. Separate observations within a cluster that have non-intersecting year-of-birth ranges, two finishes on the same date at different races, more than one finish at the same race in the same year, conflicting observed gender, or finishes labeled to incompatible countries.

4. Rarity and community splits. Break up clusters whose name combinations are common enough that the high-recall merge is likely to have collapsed several distinct people, and split observations of a candidate identity that fall in disconnected race communities, since a single physical runner is unlikely to compete in entirely separate race communities.

5. Validation sweep. Re-apply the contradiction tests on the post-split clusters. The output is a runner-id assignment for every observation in the panel.

raw results → (1) rarity tables, race communities → (2) high-recall merge
→ (3) contradiction splits → (4) rarity and community splits → (5) validation sweep →
runner IDs

As a robustness check, I re-estimate the main course effects on the subsample of clusters that pass the strictest filters (rare name combinations, narrow year-of-birth windows, no high-volume aggregator flags). The resulting estimates have a Spearman correlation of approximately 0.95 with the baseline estimates (Table B5), and Table B2 shows that the cross-race validation slopes are essentially unchanged when the entity resolution's discretionary merge steps are excluded.

A.4 Time Variable Selection

Web-scraped race results sometimes contain misaligned columns, so a variable labeled "time" may correspond to gun time, chip time, or an unrelated field. For each race-year, I therefore select the time variable that minimizes within-runner mean squared prediction error, using the runner's times in other races as the prediction benchmark. Any residual mismatch between chip and gun time is small relative to the interquartile range of estimated course effects (11 minutes).

A.5 Near-Threshold Bandwidths

Different exercises use deliberately different "near-BQ" bandwidths, matched to each design:

Table A3: Near-threshold bandwidth conventions, by exercise

Exercise	Bandwidth	Rationale
Bunching density	±5 min of posted standard	Spike width at the kink
Era sorting comparison	±5 min of neutral margin	Symmetric around the target
Dissimilarity / excess sorting	±5% of standard (neutral)	Proportional across demographics
Inframarginal control group	> 15% slower (neutral)	Course choice cannot affect Q_i
Clean-index re-estimate	Excludes ±5 min of posted standard	Index identified off non-strategic attempts
Robustness exclusion spec	Excludes ±10 min of posted standard	Identification far from the effort margin
Over-attraction diagnostic	±3% of standard (neutral)	Marginal-runner share by course
Decomposition inframarginal baseline	> 20 min outside (neutral; 15/30 variants)	No-incentive pool for channel accounting

Appendix B Estimating and Validating Course Effects

B.1 AKM Estimator

I estimate equation (1) of the main text, reproduced here for reference:

$$\ln T_{ict} = \alpha_i + \delta_c + \gamma_t + f(\text{age}_{it}, \text{male}_i) + g(W_{ct}) + \varepsilon_{ict} \tag{4}$$

where T_{ict} is the finish time (in seconds) of runner i on course c in year t, α_i is a runner fixed effect capturing persistent baseline performance, δ_c is a course fixed effect capturing the persistent speed of course c (including its typical climate), and γ_t is a year fixed effect capturing common trends across all races. The function $f(\cdot)$ is a quartic polynomial in age, centered at 40, fully interacted with gender. The term $g(W_{ct})$ is a quadratic in demeaned race-day temperature, $g(W_{ct}) = \beta_1 \widetilde{W}_{ct} + \beta_2 \widetilde{W}_{ct}^2$, where $\widetilde{W}_{ct}$ is the deviation of race-day morning temperature from the marathon's historical mean. Because mean temperature is absorbed by the course fixed effect δ_c, the weather controls capture only transitory race-year temperature shocks. (Humidity, wind, and precipitation add little explanatory power on average and are omitted from the model; they can matter at specific exposed courses, which is visible in the within-course stability diagnostic of Section B.6.)

To reduce false links in the entity resolution procedure, I define runner identities separately within broad geographic regions (U.S., Canada, UK/Ireland, Australia/New Zealand, and Other), so the same physical runner competing in both the U.S. and UK is assigned separate fixed effects in each region. The model is estimated using `reghdfe` (Correia, 2017), with standard errors clustered by regional runner.

The model assumes additive separability: after accounting for runner ability and observables, courses shift log finish times by a common amount. This is a plausible approximation for road marathons, where features such as elevation profile, terrain, and typical weather conditions tend to affect all runners proportionally in time rather than creating highly runner-specific match effects. Sections B.6 and B.7 report two diagnostic exercises in the spirit of Card, Heining and Kline (2013): within-course year-to-year stability of course effects, and a nonseparability test. Additive separability (Section 1 in the main text; regression detail in Section B.3; $\hat{\beta} \approx 0.851$ for Boston-destination and 0.933 for general cross-race pairs) supplies the mover-style cross-venue validation that is typically the third CHK diagnostic. Taken together, these exercises suggest that the additive structure is a useful approximation, though not an exact description of every runner-course combination.

The overall R^2 of equation (4) is 0.816 on 11,142,687 observations. Persistent runner differences account for the bulk of that fit; the race fixed effects contribute a modest additional share that is nevertheless the object of interest, and year effects and weather controls add a small further increment. The weather controls are significant: a 10°F warmer-than-average race day slows finish times by 2.2%, or about 4.3 minutes at a 3:20 pace.

B.2 From Fixed Effects to a Course Speed Index

The course speed index is constructed from the estimated race fixed effects $\hat{\delta}_c$, which capture persistent differences in venue advantage after netting out runner ability, age-gender profiles, and race-year shocks. Because the model is estimated separately within each geographic region, these fixed effects are identified only up to a region-specific constant. I normalize the U.S. estimates by setting $\hat{\delta}_{\text{Boston}} = 0$ and calibrate the remaining regions using approximately 24,000 trusted bridge runners who appear in both the U.S. and another region. For each non-U.S. region, I use the median difference in runner fixed effects among bridge runners to recover a calibration constant that places all course effects on the Boston-centered U.S. scale.

I then apply Empirical Bayes shrinkage to the estimated course effects to reduce noise from sparsely connected courses. Intuitively, the procedure treats each estimated $\hat{\delta}_c$ as a noisy signal of an underlying permanent course effect and pulls imprecisely estimated values toward the overall mean, with stronger shrinkage for courses estimated with greater variance

(Morris, 1983; Andrews et al., 2008). Letting $\widehat{V}_c$ denote the estimated sampling variance of $\hat{\delta}_c$ and $\hat{\tau}^2$ the cross-course variance of the underlying effects, the shrinkage factor is

$$w_c = \frac{\hat{\tau}^2}{\hat{\tau}^2 + \widehat{V}_c},$$

so the shrunk estimate is

$$\hat{\delta}_c^{shr} = w_c \hat{\delta}_c.$$

Courses with little information are therefore pulled more strongly toward zero, while well-measured courses are left largely unchanged. I retain courses with at least 50 AKM observations after shrinkage; the final index covers 1057 courses.

Because the model is estimated in logs, the course effect $\hat{\delta}_c^{shr}$ is a proportional measure: a course with $\hat{\delta}_c^{shr} = -0.05$ is approximately 5% faster than Boston for all runners, regardless of pace. This proportional property means a given percentage advantage translates to more minutes for slower runners. For a runner on pace to finish in 3:00, a 5% course advantage is worth 9 minutes; for a 3:20 runner, 10 minutes; for a 4:00 runner, 12 minutes.

Throughout the paper, I report course effects in percentage terms and provide minute equivalents at two reference paces: 3:00, the 2025 M18–34 qualifying standard, and 3:20, which approximates the pooled qualifier finish time (median 3:19:21) across 968,601 qualifier-races in the sample. I define a "fast" course as one that is at least 2% faster than Boston—equivalent to about 3.6 minutes at a 3:00 pace or 4 minutes at a 3:20 pace—which captures 250 courses (24% of the index). I define a "very fast" course as one at least 4% faster (110 courses, 10%). I define the BQ handicap as the percentage by which the qualifying standard should be changed to equalize course difficulty; because it is proportional, a single handicap applies to all age and gender groups.

One normalization property matters for the accounting downstream. The fixed-field counterfactual in Section 2 ranks all candidates by venue-adjusted time $T_i e^{-\hat{\delta}_{r(i)}}$ and admits a fixed number; adding any constant k to every course effect rescales every adjusted time by e^{-k} and leaves the ranking—hence the admitted set, hence the displacement share—unchanged. The displacement result therefore depends only on relative course effects, and only cross-course differential contamination matters, which is what the diagnostics of Section 1 measure and what the corrected-index suite bounds. The misallocation rate against the posted standard is not invariant to k; that series is therefore always presented with its Boston-state numeraire stated, and the cross-index band serves as its robustness check.

B.3 Validation: Cross-Race Additive Separability and the Boston Retest

The estimated course effects are useful only if they capture real, additively separable differences in venue speed. The cleanest test is whether the index is predictive of subsequent Boston race performance. I then generalize to the full set of cross-race pairs as a broader check.

The core specification, applied throughout this subsection, is:

$$\ln T_{i,\text{dest}} - \ln T_{i,\text{orig}} = \beta \cdot (\hat{\delta}_{\text{dest}} - \hat{\delta}_{\text{orig}}) + \mathbf{X}_i'\boldsymbol{\phi} + \varepsilon_i \tag{5}$$

where $\hat{\delta}_{\text{dest}} - \hat{\delta}_{\text{orig}}$ is the difference in estimated course effects between the destination and origin races and $\mathbf{X}_i$ collects a quadratic in age and gender. By differencing finish times within runner, this specification absorbs time-invariant runner ability. Perfect additive separability implies $\beta = 1$: moving to a course that is 1 log point faster should improve finish time by 1 log point on average. Standard errors are clustered by runner.

B.3.1 Boston as a Standardized Retest

Because Boston is the normalization point of the course index, $\hat{\delta}_{\text{Boston}} = 0$ by construction. I construct the retest sample by taking every runner-year in which the runner ran Boston and pairing it with each of that runner's marathons in the 18 months before Boston at a different course, yielding 690,058 pairs. On average, runners are slower at Boston than at their prior race, and faster qualifying courses predict larger Boston slowdowns (Figure 2, Panel A).

Table B1, column 2, reports estimates of equation (5) on the Boston-destination subsample. The estimated additive separability coefficient is $\hat{\beta} = 0.851$: for each 1% increase in a prior marathon's speed, a runner is predicted to run Boston 0.851% slower. On a raw basis, runners are on average slower at Boston than at their prior race; most of this is explained by course-FE additive separability, leaving a positive constant that captures residual Boston-specific factors such as fatigue, travel, race-day conditions, and the order/peaking behavior measured directly in Appendix C. (Figure 2, Panel A, plots the retest binscatter; Panel B plots the directional order/peaking diagnostic.)

B.3.2 General Cross-Race Additive Separability

To show that additive separability is not specific to Boston, I generalize the pair construction to all destinations. For every runner-race observation in the dataset, I pair it with each prior marathon by the same runner in the 18 months before at a different course.[4] This yields 10,934,525 pairs; Table B1, column 1, reports $\hat{\beta} = 0.933$. Column 4 estimates an interacted model that allows the slope to differ between Boston and non-Boston destinations; both the Boston-destination (essentially the column 2 slope) and non-Boston-destination ($\hat{\beta} = 0.934$) coefficients closely approach the 1.0 benchmark of perfect additive separability.

[4]The same-course restriction excludes pairs like Boston 2019 / Boston 2020, which would contribute a degenerate zero FE-difference observation; the runner still enters the analysis through cross-course pairs (e.g., Boston 2019 / NYC 2018 and Boston 2020 / NYC 2018).

Table B1: Cross-race additive separability of course fixed effects

	(1) All pairs	(2) Boston dest.	(3) Non-Boston	(4) Interacted
FE difference	0.933***	0.851***	0.934***	
	(0.002)	(0.009)	(0.002)	
FE diff × Boston				0.852***
				(0.009)
FE diff × non-Boston				0.934***
				(0.002)
Boston-dest indicator				0.042***
				(0.000)
Age controls	✓	✓	✓	✓
N	10,934,525	690,058	10,244,467	10,934,525
R^2	0.156	0.070	0.161	0.162

Notes: Dependent variable is the log time gap, $\ln T_{\text{dest}} - \ln T_{\text{origin}}$. "FE difference" is the course fixed effect at the destination minus the origin. Each runner's destination race is paired with every prior marathon at a different course within 18 months; pair-weighted WLS so prolific runners do not dominate; age and gender controls. Column (4) interacts the FE difference with a Boston-destination indicator. Standard errors clustered by runner in parentheses. $^{***}p < 0.001$.

Table B2: Additive-separability slope by pair stratum: name rarity and entity-resolution link provenance

	All destinations			Boston destination		
Pair stratum	$\hat{\beta}$	(SE)	Pairs	$\hat{\beta}$	(SE)	Pairs
All pairs	0.933	(0.001)	10,934,525	0.851	(0.004)	690,058
Rare names (≤ 2 people per name)	0.895	(0.001)	6,963,864	0.774	(0.004)	482,583
Medium (3–10 people)	1.006	(0.002)	2,939,352	0.961	(0.009)	152,046
Common (> 10 people)	1.162	(0.003)	1,031,309	1.357	(0.019)	55,429
Excluding attached yob-less links	0.936	(0.001)	5,272,574	0.857	(0.005)	441,583
Excluding residence-merge links	0.915	(0.001)	4,861,866	0.810	(0.005)	387,801
Medium, all low-confidence links excluded	0.981	(0.003)	710,472	0.928	(0.014)	62,513
Rare, all low-confidence links excluded	0.876	(0.001)	2,098,419	0.741	(0.006)	208,080

Notes: WLS estimates of equation (5) within pair strata (pair-weighted, age and gender controls). Tiers classify the runner's (first name, last name, gender) cell by the number of distinct people who share it, measured as distinct birth-year clusters. "Attached yob-less links" are runner ids that mix rows with and without a birth interval (the entity resolution's NA-attach step); "residence-merge links" are ids consolidated by the modal-residence re-merge. Slopes above one among common names indicate occasional false merges—a spurious mover pairing a faster person with a faster course—while slopes into Boston sitting below slopes elsewhere indicate origin-race peaking (Appendix C). Excluding the entity-resolution merge steps leaves the slopes essentially unchanged, so the deviations are not artifacts of the linkage design.

B.3.3 Course-Level Additivity Residuals

For each origin course c with at least 50 runner-pairs (i, c, Boston) where the same runner raced the origin course and Boston within 18 months, I compute the average log-time gap and compare it to the FE-implied gap. Let $g_{ic}^{\text{obs}} = \ln T_{i,\text{Boston}} - \ln T_{ic}$ denote the observed log-time difference (Boston minus origin) for runner i's pair and $g_{ic}^{\text{FE}} = \hat{\delta}_{\text{Boston}} - \hat{\delta}_c = -\hat{\delta}_c$ the corresponding FE-implied difference (since $\hat{\delta}_{\text{Boston}} = 0$ by normalization). If the AKM is perfectly additively separable, the expected value of g_{ic}^{obs} is g_{ic}^{FE} and the residual is mean-zero. The course-level additivity residual is the course-specific average of that residual, after partialling out year-to-year temperature deviations at the origin and at Boston:

$$\widehat{\Delta}_c \;=\; \mathbb{E}_{i\in\mathcal{P}_c}\Big[\big(\ln T_{i,\text{Boston}} - \ln T_{ic}\big) \;-\; \big(-\hat{\delta}_c\big)\Big] \;=\; \mathbb{E}_{i\in\mathcal{P}_c}\big[\ln T_{i,\text{Boston}} - \ln T_{ic}\big] \;+\; \hat{\delta}_c, \tag{6}$$

where $\mathcal{P}_c$ is the set of runners who raced both course c and Boston within the 18-month window. Reported in minutes at a 3:20 reference pace, $\widehat{\Delta}_c$ measures how much slower (or faster) runners actually performed at Boston than the origin course's fixed effect would have predicted. A perfectly additively separable index would have $\widehat{\Delta}_c = 0$ at every course; positive values indicate the FE overstated the origin course's speed (runners were slower at Boston than the FE implied), and negative values indicate the FE understated it.

Figure B1 plots $\widehat{\Delta}_c$ against the origin-course FE for 345 courses with $|\hat{\delta}_c| \leq 25$ minutes. Most courses sit inside a ± 5 minute reference band around zero: the FE index is additively separable to within a few minutes course by course. Most of the small positive shift in the mean (+2.5 minutes) reflects a Boston-specific slowdown common to all origin courses—travel fatigue, the April race date, and the Boston course's own well-known difficulty—that the FE normalization to Boston cannot absorb at the individual-runner level. $\widehat{\Delta}_c$ is the course-level analogue of the pair-level residual η of Section 1; the course-weighted mean here (+2.5 minutes, temperature-partialled, over courses with at least 50 pairs) is smaller than the pair-weighted $+ + 7.9$ minutes of Table C1 because high-volume strategic origins with large positive residuals dominate the pair-weighted average. The labeled outliers have identifiable causes: Brooklyn's course-measurement history (alternative paths through Prospect Park in different years), heavily peaked goal-race subsamples at Mt. Hood and Clearwater, and LA's mild-weather baseline, which underweights the Boston heat slowdown. The BQ.2 courses show small negative gaps: their qualifiers ran Boston slightly faster than the origin FE predicted, matching the negative peaking residuals of Table C2—if anything, their fast fixed effects understate the speed that transfers to Boston.

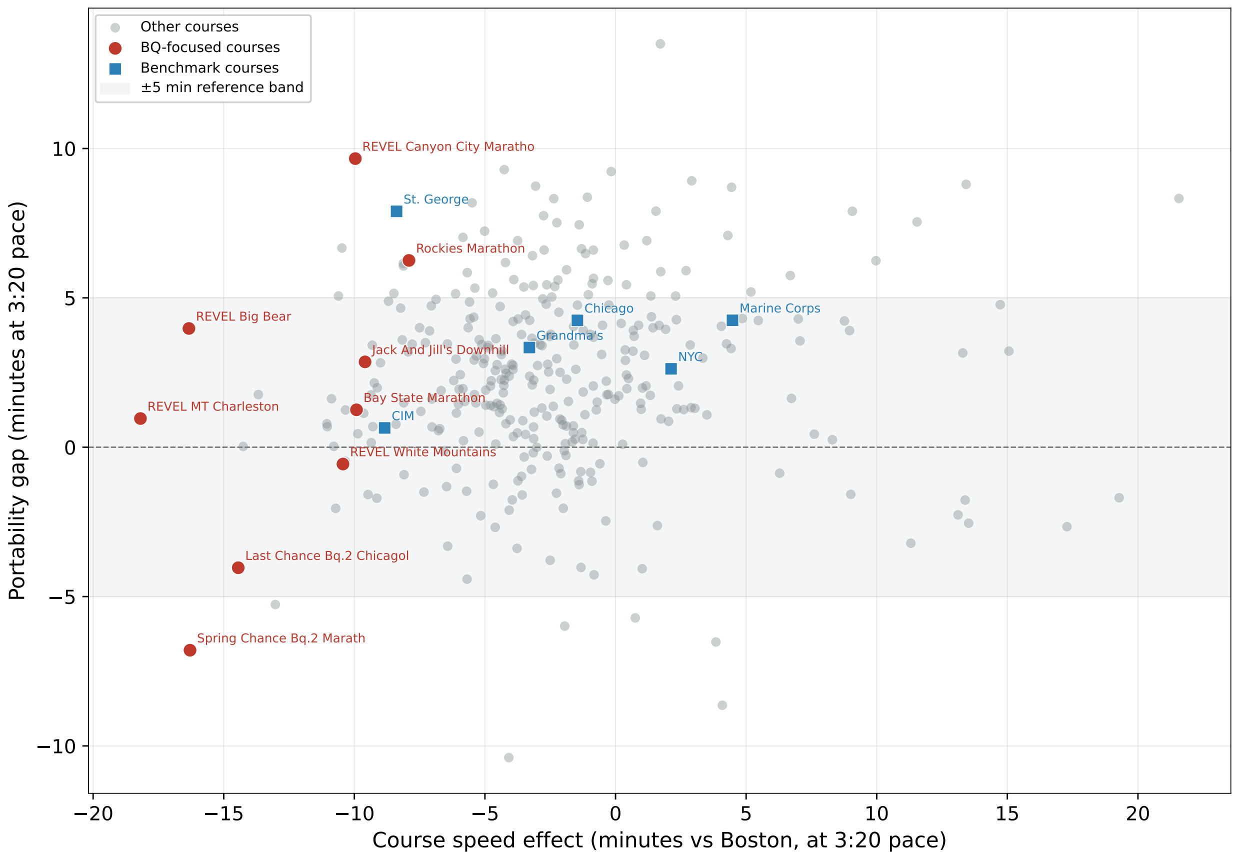

Figure B1: Course-level additivity residual $\widehat{\Delta}_c$ (equation (6)) against the origin-course FE, for qualifying courses with at least 50 same-runner pairs who subsequently raced Boston within 18 months (weather-adjusted; $|\hat{\delta}_c| \leq 25$ minutes; negative FE values are courses faster than Boston). Perfect additive separability implies $\widehat{\Delta}_c = 0$; positive values mean runners slowed more at Boston than the origin FE predicted. Most courses sit inside the ± 5 minute reference band; the mean sits at $+2.5$ minutes, a Boston-specific slowdown common to all origins.

B.4 Destination-Specific Validation

The transfer relationship is not an artifact of the Boston-bound population: Table B3 reruns the retest with other destinations, and the slopes into the lottery-entry majors bracket the Boston slope.

Table B3: Boston retest by destination numeraire: same-runner pairs into each destination; slope of the realized log-time change on the predicted course difference. SEs clustered by runner.

Destination race	Slope $\hat{\beta}$	SE	Pairs
Boston	0.851	(0.009)	690,058
Chicago	0.988	(0.035)	11,852
New York City	0.998	(0.038)	2,438
Marine Corps	0.826	(0.039)	62,607
Philadelphia	0.824	(0.046)	19,941
All destinations	0.933	(0.002)	10,934,525

B.5 Observable Predictors of Course Speed

Course fixed effects summarize everything that makes a marathon systematically faster or slower after accounting for runner ability, age, year, and realized weather. That breadth is their advantage, but it means the index does not say which course features matter. To ask how much observable course information can recover, I use the same policy-relevant universe as the main analysis: 439 profiled U.S. road courses with net and gross elevation, maximum elevation, and typical weather. I estimate

$$\widehat{\delta}_c = \mathbf{X}_c'\boldsymbol{\beta} + u_c, \tag{7}$$

where $\widehat{\delta}_c$ is the EB-shrunk log course effect and $\mathbf{X}_r$ contains observable course characteristics.

Table B4 reports both in-sample and cross-validated R^2. All columns use exactly the same courses. Net drop alone predicts little; gross climbing is substantially more informative, and combining net and gross elevation improves fit further. Adding maximum elevation and typical temperature, wind, and precipitation produces an in-sample R^2 of 0.57, but only 0.47 when predicting held-out courses. Because the AKM target is approximately 0.96 reliable, sampling error can raise that predictive fit only to 0.49. The remaining error therefore primarily reflects omitted or coarsely measured features—grade sequencing, turns and loops, route vintages, crowding, exposure, and race operations—not an imprecisely estimated target.

Table B4: Observable predictors of course speed on U.S. road courses

	(1) Net drop	(2) Gross gain	(3) Net + gross	(4) Physical + weather
Courses	439	439	439	439
In-sample R^2	0.08	0.41	0.47	0.57
10-fold cross-validated R^2	0.07	0.33	0.39	0.47

Notes: The dependent variable is the EB-shrunk AKM log course effect, normalized to zero at Boston; positive values indicate slower courses. Estimates are course weighted and use an identical sample of profiled U.S. road courses in every column. Column (4) includes net drop, gross gain, maximum elevation, typical temperature and its square, wind, and precipitation. Cross-validation randomly assigns entire courses to ten folds and predicts each fold using the other nine. Approximate repeat-sample reliability of the dependent variable is 0.96; the corresponding attenuation-adjusted column-(4) cross-validated R^2 is 0.49.

B.6 Within-Course Year-to-Year Stability

Following Card, Heining and Kline (2013), I assess the AKM decomposition on two further dimensions: year-to-year stability of course effects within each race, and nonseparability of runner and course effects. These exercises are intended as stress tests of the additive approximation rather than as alternative estimators of the main course index.

If the estimated course effect reflects a persistent feature of the race, the year-to-year variation in finish times at a given course, net of common year trends and runner composition, should be small relative to the cross-course variation in the index. I measure this stability using the standard deviation of race-year mean residuals within each course. After the AKM, for each race-year cell I compute $\bar{e}_{c,t} = \text{mean}(\hat{\varepsilon}_{i,c,t})$; the within-course across-year SD of $\bar{e}_{c,t}$

measures year-to-year course variation net of the common year FE.

Across the 691 courses with at least three years of data, the median within-course across-year SD is 2.8 minutes (at a 3:20 pace), the 95th percentile is 6.7 minutes, and the mean is 3.3 minutes. These are small compared with the cross-course range of the permanent course effect itself (more than 40 minutes), implying that $\hat{\delta}_c$ is well identified as a time-stable course attribute. Figure B2 plots this within-course across-year SD against the permanent course effect for each race, highlighting benchmark courses. Marine Corps (1.9 min), St. George (2.2 min), and Honolulu (2.2 min) are among the most stable despite spanning a wide range of course speeds; Boston itself is at 3.7 minutes, reflecting genuine year-to-year variation in weather and field composition. The remaining labeled high-SD outliers have thin samples or weather-exposed designs, not identification failures: Empirical Bayes shrinkage has already pulled them toward zero.

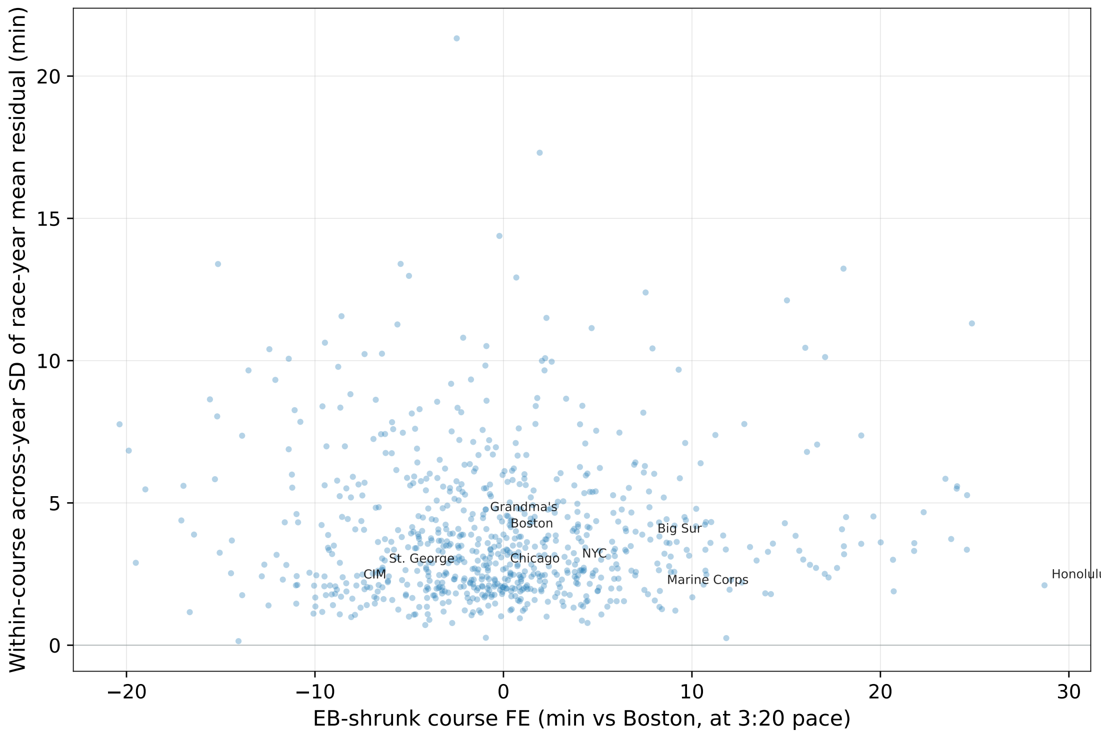

Figure B2: Within-course year-to-year stability of the AKM course effect. Horizontal axis: EB-shrunk course FE in minutes vs Boston at 3:20 pace (negative values are faster than Boston; $|\hat{\delta}_c^{shr}| \leq 25$ min plus benchmarks always retained). Vertical axis: standard deviation of race-year mean residuals within that course, net of the year, runner, and course effects, the age-gender polynomial, and temperature deviations. The median across-year SD is 2.8 minutes and the 95th percentile 6.7 minutes, small relative to the more-than-40-minute cross-course range.

B.7 Nonseparability

Under the additive AKM structure, the course effect shifts all runners' log times by the same amount, so mean residuals should be approximately zero in every runner-ability $\times$ course-speed cell. To test this, I assign each observation to a runner-FE decile and a course-FE decile and compute the mean residual in each of the 100 cells. I then two-way demean to isolate the runner $\times$ course interaction: $\hat{m}_{rc} = \bar{e}_{rc} - \bar{e}_{r\cdot} - \bar{e}_{\cdot c} + \bar{e}_{\cdot\cdot}$, where r indexes runner-FE deciles and c course-FE deciles. Under the additive model, $\hat{m}_{rc} \approx 0$ for all cells.

Figure B3 displays the interaction heatmap. The largest absolute cell deviation is 1.11 log points $\times$ 100 (approximately 1.1% of finish time), in the extreme corner cell pairing the slowest runner decile with the fastest course decile, and the mean absolute deviation is 0.18%. The interaction terms do exhibit some systematic structure, but it is concentrated in extreme runner-course cells and modest relative to the main course effects. In particular, the largest deviations occur in the tails of the runner and course distributions rather than in the middle of the support where the BQ-margin analysis is concentrated.

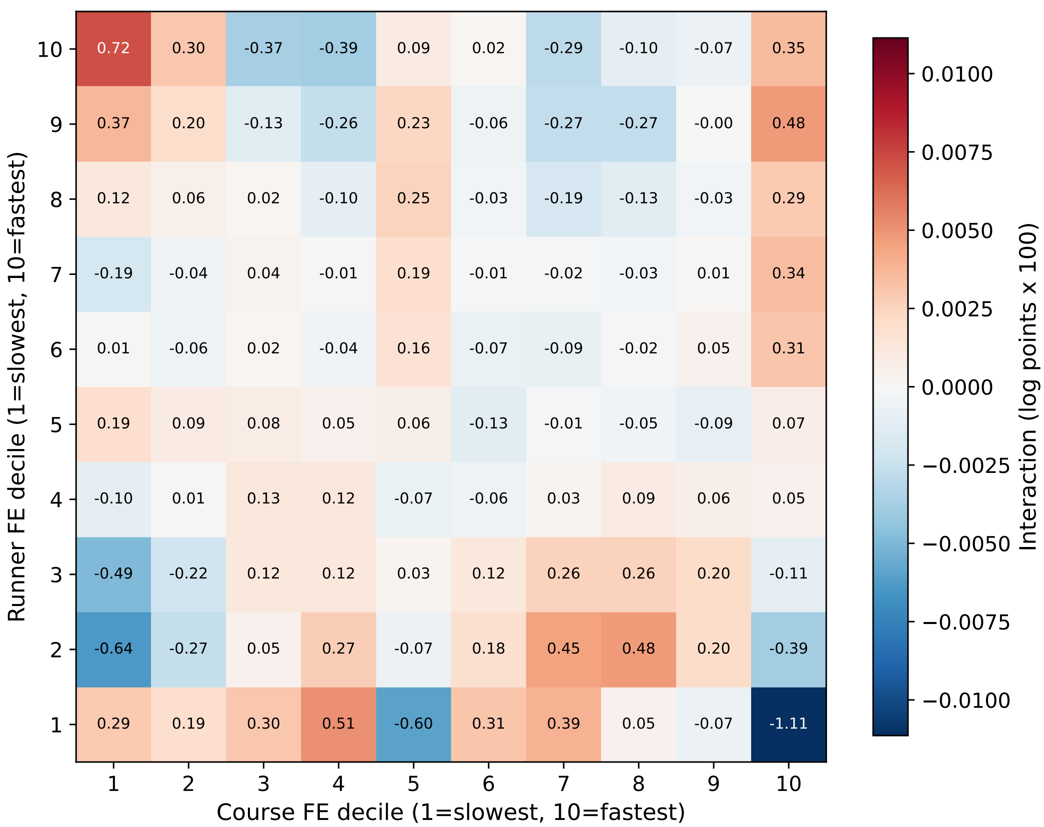

Figure B3: Nonseparability test: two-way demeaned interaction residuals by runner-FE decile (rows) and course-FE decile (columns), each ordered from slowest to fastest. Values are log points $\times$ 100 (approximately percent). Under the additive AKM model, all cells should be near zero.

B.8 Sample-Restriction Robustness

I estimate six alternative specifications. The first three alter the sample restrictions: excluding trail races, restricting to high-confidence entity-resolution matches, and requiring at least three appearances per runner. The fourth drops observations within ± 10 minutes of the BQ standard, isolating identification from runners far from the strategic-effort margin. The fifth replaces each finish time with its debunched value (subtracting the mass piled up at salient minute marks within each race-year). The sixth restricts to runners who placed top 5 in their age-gender group at any race—"race-to-win" finishers for whom BQ pursuit is not the operative incentive. Four of the six specifications yield Spearman rank correlations of 0.99 or higher with the baseline estimates (Table B5). The two that deviate most are the high-confidence entity-resolution restriction (Spearman 0.953, mean $|\Delta\text{FE}| = 2.3$ minutes), which discards most of the mover network, and the top-5 specification (0.969, 2.1 minutes), consistent with the elite tail of finishers being a different selection of runners than the BQ-pursuing population that dominates the baseline sample.

Table B5: Spearman rank correlations with baseline course fixed effects across robustness specifications. Mean $|\Delta\text{FE}|$ reports the average absolute change in estimated course speed (minutes at 3:20 pace) relative to the baseline.

Specification	N Races	Spearman ρ	Mean $|\Delta\text{FE}|$ (min)
Sample restrictions			
Exclude trail races	730	0.999	0.2
High-confidence ER only	721	0.953	2.3
Min 3 appearances	799	0.998	0.4
Strategic-effort robustness			
Excl. BQ margin ± 10 min	804	0.994	0.8
Debunched finish times			
Debunched (medium $[-5, +7]$)	815	1.000	0.1
Non-marginal runners			
Top 5 by age group	804	0.969	2.1

Appendix C Strategic Effort and the Corrected Course Index

This appendix collects the material behind the behavior-adjusted index of Section 1: the decomposition of the naive course effect, the ordered edge/peaking diagnostics, the strategic-share and behavior-adjusted-index detail, the construction of the routine-attempt and reverse-pair estimators, the implementable conservative and observable-predicted variants, the consolidated index suite, and robustness of the AKM index itself to strategic-effort contamination.

What an AKM regression recovers for each course is the observed venue value-added,

$$\hat{\delta}_c^{\text{naive}} = \underbrace{\delta_c}_{\text{systematic course speed}} + \underbrace{S_c}_{\text{selection into venue}} + \underbrace{E_c}_{\text{target effort / peaking}} + \underbrace{O_c}_{\text{order / edge effects}} + \text{noise}, \tag{8}$$

because the runner effect absorbs each runner's average state but not the systematic within-runner effort, peaking, and ordering that correlate with venue choice. The term $s_c b_c$ of equation (2) collects $S_c + E_c + O_c$; Section 1 measures the composite and the diagnostics below sign its components.

C.1 Ordered-Pair Residual Estimates

Table C1: Ordered edge/order decomposition of the additivity residual η (minutes at 3:20 pace).

	(1) Order	(2) +Near-BQ	(3) +Goal/qual.
Destination is Boston	7.930***	7.400***	2.907***
	(0.060)	(0.072)	(0.105)
Origin is Boston	-2.134***	-3.326***	-6.374***
	(0.087)	(0.088)	(0.089)
Near BQ at origin		9.872***	5.424***
		(0.082)	(0.087)
Dest. Boston × near BQ at origin		-6.076***	-1.601***
		(0.140)	(0.144)
Qualified at origin			12.488***
			(0.076)
Dest. Boston × qualified at origin			-3.246***
			(0.134)
Major-city destination			0.296***
			(0.046)
Temperature deviation difference (°F)	0.344***	0.340***	0.333***
	(0.002)	(0.002)	(0.002)
Gap, age, temperature controls	Yes	Yes	Yes
Ordered pairs	10,934,525	10,934,525	10,934,525
Runners	1,331,603	1,331,603	1,331,603

Notes: Pair-weighted WLS on ordered same-runner pairs; the dependent variable is the additivity residual η in minutes at a 3:20 pace, and positive values mean the runner was slower at the destination than the additive model predicts. Columns add controls progressively; every specification controls for the months between the paired races, the age change, and the temperature-deviation difference. "Major-city destination" is an indicator for a destination among Chicago, New York City, Marine Corps, Houston, Philadelphia, Grandma's, Twin Cities, or Boston. Standard errors clustered by runner in parentheses; two-way runner × origin-course standard errors are similar and reported in the replication archive. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$.

Averaging the $A \to B$ and $B \to A$ directions of same-pair gaps differences out order and peaking components that are direction-specific, identifying $\delta_B - \delta_A$ under the assumption that order effects are symmetric in expectation. The asymmetry it averages out is the one shown in Figure 2, Panel B; the resulting reverse-pair estimator appears in the suite of Table C3.

C.2 Strategic Shares and the Adjusted Index

Table C2: Strategic share and the behavior-adjusted index for the fastest courses.

Course	Naive adv. (min)	Course-dep. qual. (%) of finishers	Course-dep. qual. (%) of qualifiers	Retest η (min)	Adjusted adv. (min)
REVEL MT Charleston	-17.1	27.3	76.5	6.3	-15.5
Spring Chance BQ.2 Marathon	-16.6	47.4	88.5	-3.4	-18.1
Tunnel Lite Marathon	-15.6	23.9	62.3	0.8	-15.4
Last Chance BQ.2 Chicagoland Marathon	-15.3	46.0	82.8	-1.4	-15.9
REVEL Big Bear and Half Marathon	-15.1	22.6	64.1	7.7	-13.6
Last Chance BQ.2 Grand Rapids Marathon	-14.5	47.1	85.3	-4.5	-16.4
Light At The End Of The Tunnel Marathon	-12.8	24.9	69.9	6.2	-11.4
MT. Nebo Marathon	-11.4	10.0	51.6	6.7	-10.8
Hudson Mohawk Marathon	-11.2	18.2	53.6	7.1	-10.0
Carmel Marathon	-11.2	11.6	51.1	7.7	-10.4
Loco Marathon	-10.0	14.7	56.4	4.2	-9.4
Tunnel Vision Marathon	-9.9	19.8	57.9	2.2	-9.5
REVEL Canyon City Marathon	-9.6	12.4	51.1	16.1	-7.8
Bay State Marathon	-9.6	18.7	54.8	10.6	-7.7

Notes: "Course-dependent qualifiers" are performances that clear the posted standard only by virtue of the course; the two share columns express them as a percent of all finishers (s_c, the strategic share) and as a percent of apparent qualifiers. The retest residual η is the Boston-retest peaking residual: positive values mean the naive estimate makes the course look too fast. "Adjusted advantage" is equation (3); in both advantage columns negative values are faster than Boston. Because the correction adds $s_c\,\eta_c$, a negative residual makes a course's adjusted advantage larger, not smaller: those courses' qualifiers ran Boston faster than the additive model predicted, so the naive index understates the speed that transfers.

C.3 Routine-Attempt Estimator

A natural alternative to correcting the pooled index is to split the sample and estimate the model separately for targeting and non-targeting attempts. Targeting, however, is a runner-race state, not a runner type: the same person targets in some attempts and not others, so the pair, not the runner, is the right unit of a split. The routine-attempt estimator implements that split. An attempt is routine ex ante if the runner's geocoded home is within 100 miles of the course or the runner has run that course in a prior year: locals and repeaters run Chicago because it is their race, not as a once-peaked goal. Estimating course effects from routine–routine same-runner pairs only (a within-runner pair estimator on 1.9 million pairs) gives a routine-only index whose correlation with the naive index is 0.96 across 733 courses. Each course's visitor premium—naive minus routine-only, after removing the common Boston-numeraire level that the invariance result (Section B.2) makes irrelevant—measures how much of its apparent speed is visitor composition. The premium concentrates where peaking predicts: dedicated goal venues are flattered by visitor composition, ordinary majors are not, and the interquartile range across courses is -2.7 to +1.5 minutes (the course-level premia are tabulated in the replication archive). The contamination has the predicted sign, can be attributed course by course, and is bounded. The split fails only where targeting is the product itself: no one runs a last-chance qualifier casually. The corrected-index suite therefore pairs it with corrections that need no non-targeting subsample.

C.4 Conservative and Observable-Based Variants

Two implementable variants address estimation error. A confidence-adjusted handicap shrinks each course's adjustment toward zero by k standard errors, zeroing courses statistically indistinguishable from Boston; at $k = 1.645$ displacement falls to 10.2%. An observable-predicted rule uses predicted speed from terrain, surface, and climate—applicable to brand new courses with no race history—and yields 9.0%; it under-adjusts the fast-flat loop courses whose speed does not come from elevation, which is also why it is not a behavior-free benchmark either. A practical implementation uses the observable formula for new courses and the estimated index for established ones, updating annually.

C.5 The Consolidated Index Suite

Table C3: Alternative course-effect indices: construction, the bias each addresses, coverage, agreement with the headline index, and the implied fixed-field displacement (US 2024–25 qualifiers, field size 87,066).

Index	Construction	Bias addressed	Courses covered	Rank corr. w/ headline	Displaced (%)
Behavior-adjusted (headline)	Naive AKM $+s_c\,\eta_c$, eq. (3)	Course-level peaking of strategic users	1,057	1.00	11.9
Naive AKM (starting point)	EB-shrunk course fixed effect	—	1,057	1.00	12.5
Clean re-estimate	AKM excluding ±5 min of the standard	Strategic effort near the threshold	815	1.00	12.3
Routine-attempt	Local/repeat same-runner pairs only	Visitor/goal-race composition	734	0.84	10.4
Reverse-pair (symmetric)	Averages $A\to B$ and $B\to A$ gaps	Direction-specific order effects	1,057	0.92	12.2
Observable-predicted	Terrain, surface, climate formula	All non-physical variation (and some physical)	531	0.70	9.0
Conservative shrinkage (k=1.645)	Zeroes courses within 1.645 SE of Boston	Estimation noise	357	0.96	10.2

Notes: Rank correlations are Spearman correlations with the behavior-adjusted headline index over each variant's covered courses. Displacement is the one-sided share of the fixed field reallocated when candidates are ranked by the index-adjusted margin. Policy rules (the BAA net-drop penalties and the symmetric elevation formula) answer a different question—how a feasible rule performs against the benchmark—and are compared in Appendix Table F1.

C.6 Robustness to Strategic-Effort Contamination

A concern for the behavioral estimates of Section 3—and for any use of the runner fixed effects as neutral-course baselines—is that the runner fixed effects may themselves be contaminated by strategic effort near the BQ threshold. If runners systematically exert more effort when close to qualifying, their estimated baseline performance could be biased upward, which would in turn bias any residual built on it. I address this concern in two ways.

First, I re-estimate the AKM model after excluding runners within 5 or 10 minutes of the BQ cutoff. I also estimate the model on a sample restricted to far-from-threshold runners, defined as those whose predicted baseline performance is more than 5 minutes slower than the BQ threshold on a neutral course. In this sample, proximity to the threshold cannot affect effort. In all three cases, the estimated course effects are substantively unchanged relative to the baseline, indicating that the course-speed index is not driven by strategic effort.

Second, I re-estimate the full AKM specification after dropping all observations within ± 5 minutes of the BQ cutoff. This produces "clean" runner fixed effects identified only from observations for which runners had no immediate BQ incentive to exert differential effort. These fixed effects are highly correlated with the baseline estimates, indicating that runner-level neutral-course baselines are not sensitive to strategic-effort contamination.

A further concern is that strategic bunching at BQ thresholds and round numbers mechanically inflates course fixed effects at races where bunching is concentrated. I address this by estimating a smooth counterfactual finish-time density outside an excluded bunching region around each threshold, identifying excess mass, and probabilistically reallocating bunching runners across the excluded region according to the counterfactual density. I run three asymmetric excluded windows as sensitivity: narrow $[-3, +5]$ minutes, medium $[-5, +7]$ minutes, and wide $[-7, +10]$ minutes. The debunched course effects are virtually identical to the baseline (Spearman ≥ 0.9997 for all three windows), indicating that bunching does not materially affect the estimated course speed index.

Finally, to test whether course effects are driven by BQ-motivated effort, I identify runners who placed in the top K of their age-gender group at any race—runners who compete to win, not to qualify for Boston—and keep all their observations across all courses. The resulting course effects are highly correlated with the baseline (Spearman 0.969 for the top-5 variant; Table B5) and, if anything, show BQ-focused courses as faster, not slower, than the baseline estimates. This is inconsistent with BQ-related effort inflating the course fixed effects.

Appendix D Qualification and Field Reallocation

D.1 Robustness of the Course-Dependence Series

The performance-level headline counts race-results, so a runner with multiple qualifying races can contribute more than once. Figure D1 therefore collapses to one runner–cohort observation. A unique runner is course-dependent only if the runner has an observed posted-standard qualifier and no observed Boston-equivalent posted-standard qualifier anywhere in the window; the earlier "any course-dependent race" rule is not used. The broader sample treats unindexed-course qualifiers as unclassified rather than using them to establish participant course dependence. The entrant-level effective- cutoff series in the main figure is stricter still and is the statistic used for claims about the realized Boston field.

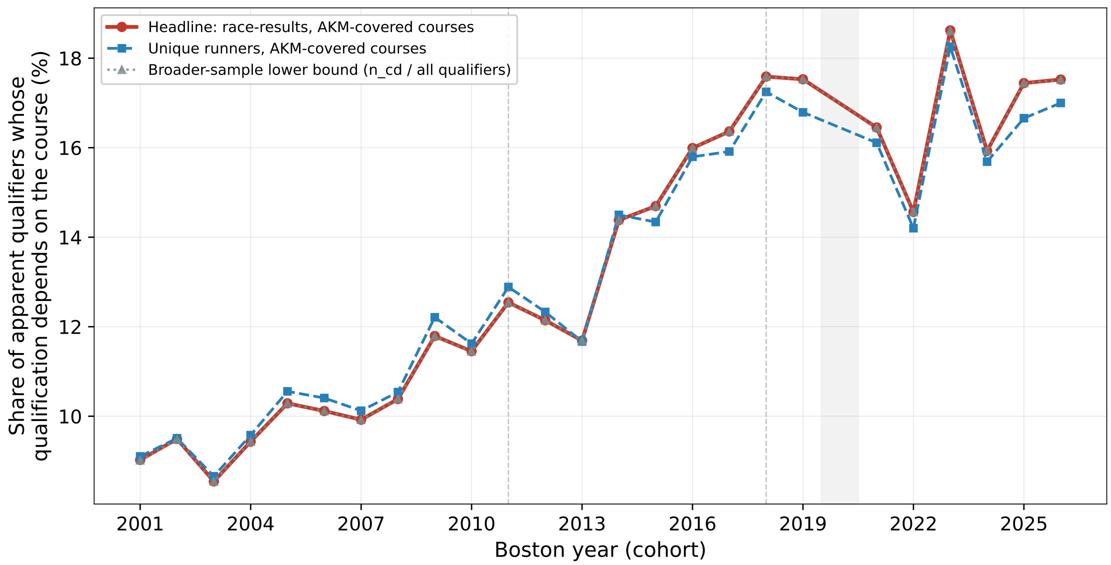

Figure D1: Robustness of the performance-level course-dependence series. The vertical axis is the share of apparent qualifiers whose qualification depends on the course. The unique-runner series counts a runner as course-dependent only when none of the runner's observed qualifying-window performances is a Boston-equivalent posted-standard qualifier. The broader-sample line expands the denominator to qualifiers at unindexed courses, which remain unclassified. Boston 2000 is excluded; Boston 2020 is shaded; dashed lines mark standard changes.

D.2 Qualifying Mechanics

Two mechanical facts explain why the distortion grows with every tightening. Each runner faces a menu of courses at which their ability can clear the standard, and the menu contracts sharply as the standard tightens: a runner with a neutral-course baseline of 195 minutes could qualify at 196 indexed courses under the 3:10 standard, 52 under 3:05, and 12 under 3:00 (Figure D2, Panel A). And tightening is not neutral across venues: on slow courses the cutoff sits in the thin right tail of the ability distribution and a five-minute tightening disqualifies few runners; on fast courses (where qualification rates run roughly 35–56% at the flagship purpose-built products, several times the 10–12% rates at the majors) it crosses dense regions and disqualifies many (Panel B), so the fast-course share of qualifiers rises mechanically with every tightening, before any behavior.

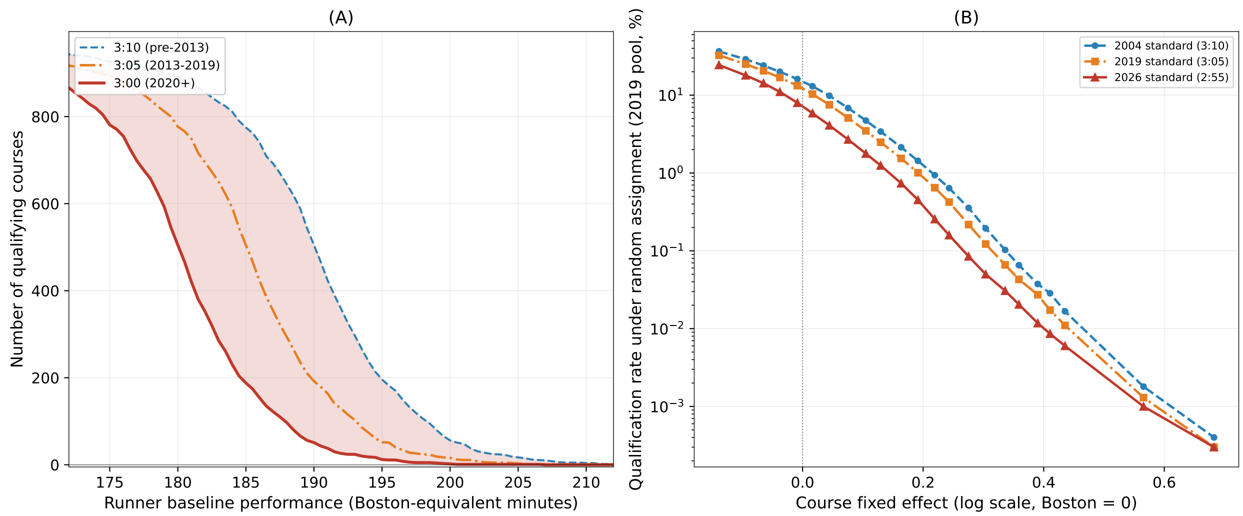

Figure D2: Qualifying mechanics. Panel A: the number of indexed marathons at which a runner of given neutral ability can qualify, under three historical standards. Panel B: qualification rates by course speed under the 2004, 2019, and 2026 standards (3:10, 3:05, 2:55). Course speed is the log course effect, with negative values faster than Boston; each 0.01 log point is about two minutes at a 3:20 pace.

D.3 Predicted Boston Performance of Displaced and Replacement Qualifiers

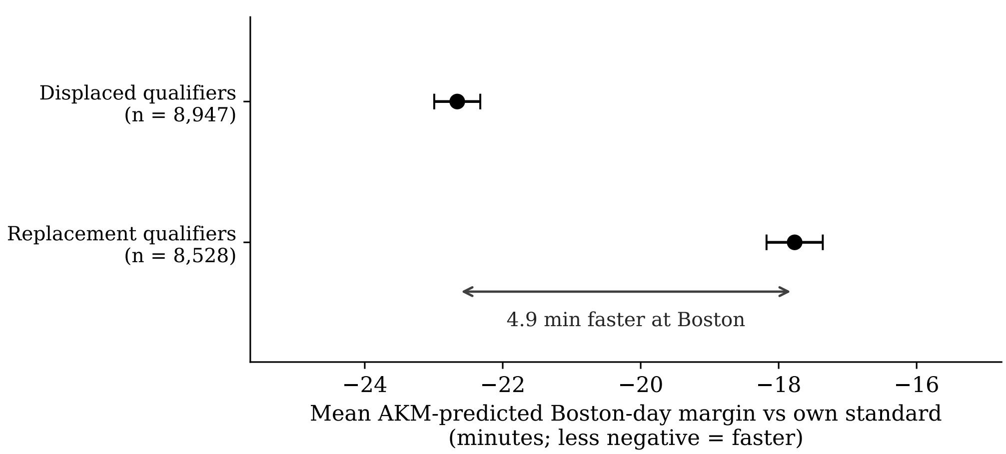

Figure D3: Predicted Boston-day performance of displaced vs. replacement qualifiers. Dots are group means of the AKM-predicted Boston-day margin (runner effect plus age–gender profile, Boston course effect zero) relative to each runner's own standard, with 95% confidence intervals. Both groups are predicted to run slower than their standards at Boston, so both means are negative; the result is the gap between them, labeled directly: the replacements are predicted 4.9 minutes faster at Boston than the runners they displace.

D.4 Decomposition Chains and Alternative Baselines

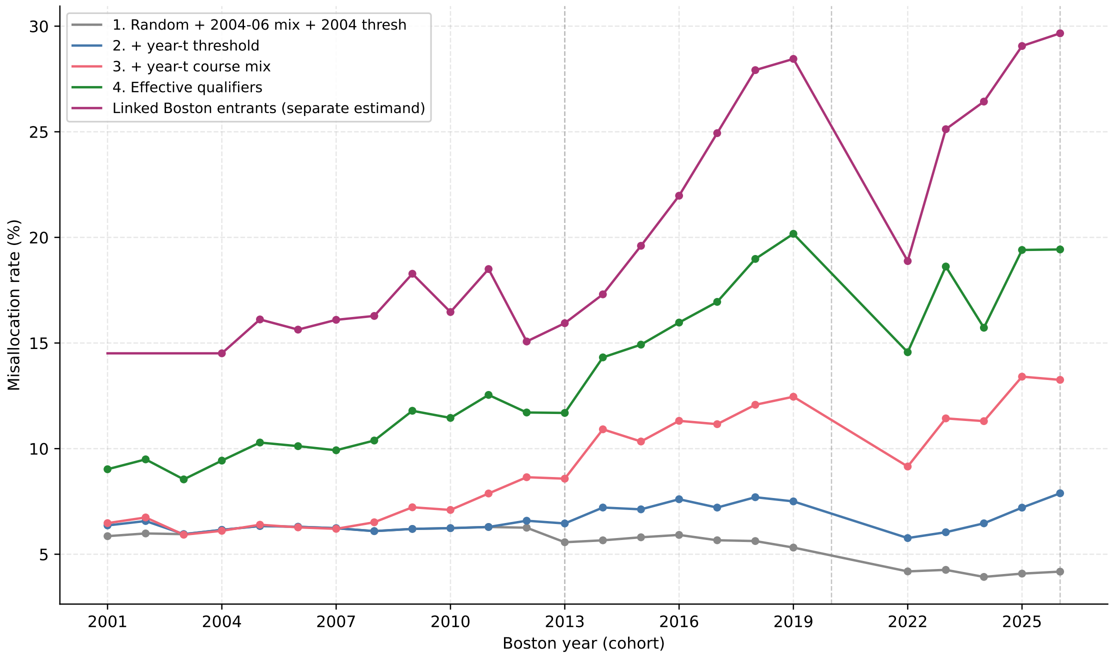

Figure D4: Counterfactual accounting of the misallocation rate, Boston 2001–2026 cohorts. Steps 1–4 add the effective cutoff, course menu, and actual assignment to the random baseline. The separate linked-entrant line reports the direct Boston-field measure and is not an additive take-up channel. Vertical lines mark the first cohort under each new standard.

Table D1: Channel contributions to the 2025 misallocation rate (pp) under alternative assignment baselines and path orderings. Sorting is invariant to the ordering; reordering reallocates only between the threshold and supply channels. Residence-subsample rows are comparable to each other, not to the full-sample rows.

Baseline	Ordering	Threshold	Supply	Sorting
Random (field shares)	Threshold first	3.1	6.2	6.0
Random (field shares)	Menu first	5.5	3.8	6.0
Inframarginal ($\lvert$margin$\rvert > 20$ min)	Threshold first	2.9	5.5	7.2
Inframarginal ($\lvert$margin$\rvert > 20$ min)	Menu first	5.1	3.4	7.2
Inframarginal ($\lvert$margin$\rvert > 15$ min)	Threshold first	3.0	5.6	7.0
Inframarginal ($\lvert$margin$\rvert > 30$ min)	Threshold first	2.8	5.4	7.5
Inframarginal, gender-conditioned	Threshold first	2.9	5.5	7.2
Random, residence subsample	Threshold first	3.4	6.2	8.2
Gravity (distance decay), residence subsample	Threshold first	3.3	5.4	9.6
Gravity (distance decay), residence subsample	Menu first	5.6	3.0	9.6

Appendix E Runner and Venue Responses

E.1 Sorting Time Series

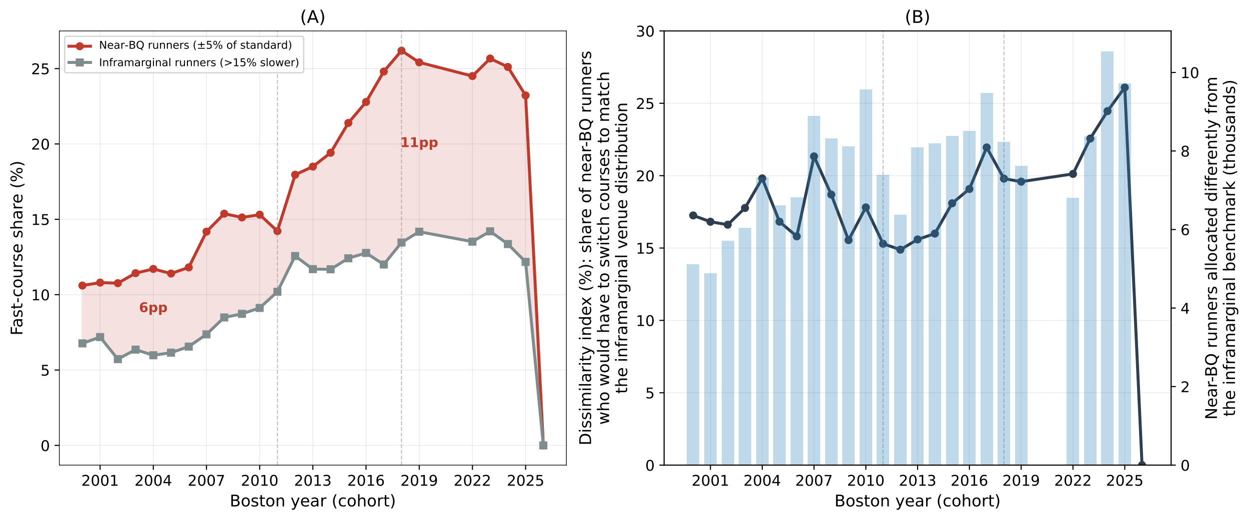

Figure E1: Near-BQ vs. inframarginal fast-course shares (Panel A) and the implied excess sorting (Panel B), 2000–2025. The exercise is descriptive: it treats the inframarginal runners' venue distribution as the no-incentive counterfactual. Under that assumption, roughly one in five near-BQ runners is at a course they would not otherwise have chosen. The dissimilarity index is the share of near-BQ runners who would have to switch courses to match the inframarginal venue distribution; the bars count near-BQ runners allocated differently from the inframarginal benchmark.

E.2 Round-Number Placebo

Figure E2 provides additional detail on the contrast between generic round-number bunching and BQ-specific sorting. The left panel examines the 4:00 mark (excluding runners for whom 240 minutes is a BQ threshold); the right panel examines the runner's own BQ threshold. The fast-course share is essentially flat across 4:00, whereas it rises sharply approaching the BQ threshold. The moving-BQ exercise in the main text (Figure 5) shows the corresponding density evidence.

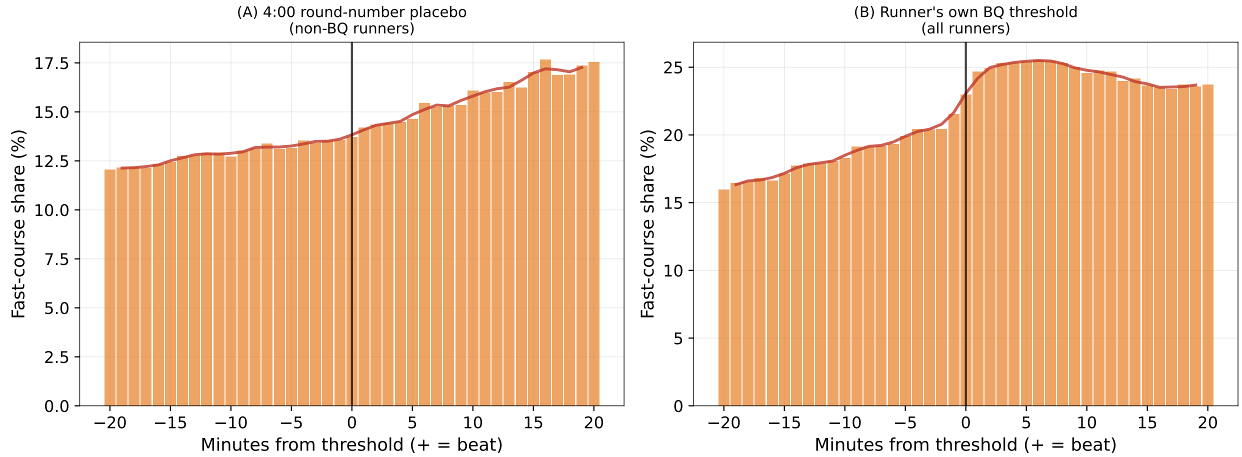

Figure E2: Fast-course share around a generic round number and around the BQ threshold. The left panel examines the 4:00 mark, excluding runners for whom 240 minutes is the BQ threshold; the right panel examines the runner's own BQ threshold. Sorting into fast courses is limited at the generic round number but pronounced at the BQ cutoff.

E.3 Entry of Fast Courses

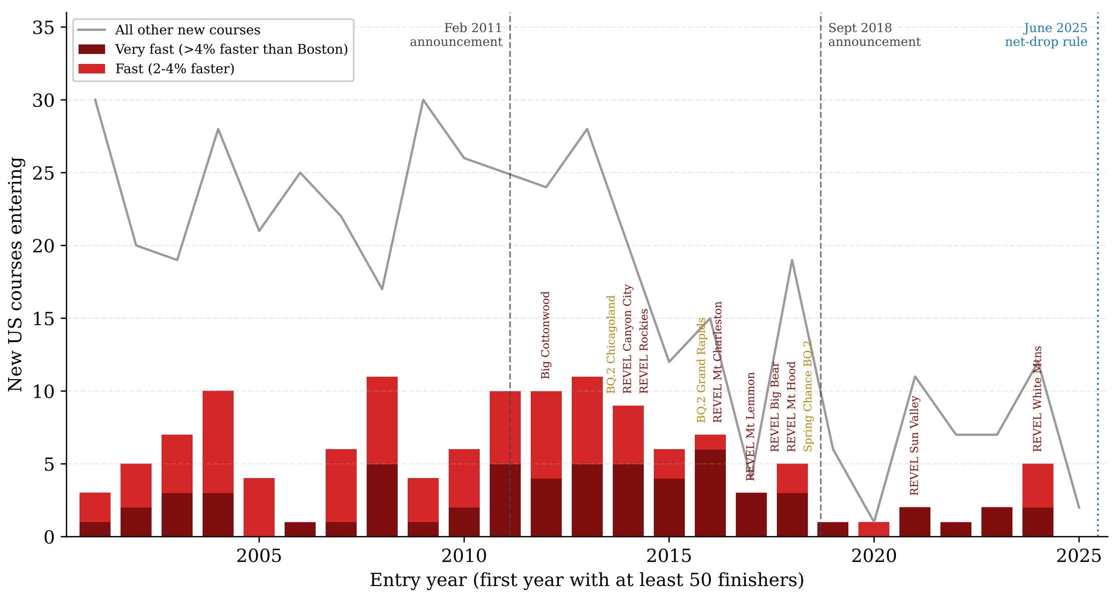

Figure E3: Fast-course entry and the launch of purpose-built BQ races. Entry of US road marathon courses by course-speed class, 2001–2025, with dedicated qualifying products labeled by name; dashed lines mark the February 2011 and September 2018 announcements and the June 2025 net-drop rule. Entry year is a course's first panel year with at least 50 finishers; courses already active in 2000 are left-censored incumbents and excluded.

E.4 Take-Up

Boston admits a fixed field from qualified applicants, so observed appearance combines application, acceptance, attendance, and successful linkage. I measure take-up only among performances clearing the effective admission cutoff, not among everyone who merely clears the posted application standard. Across cohorts the resulting observed Boston-appearance rate is about 31.4%. Entity-resolution failure can still deflate the level and can bias cross-course comparisons if match rates differ by course. Restricting to linkage-homogeneous US-resident effective qualifiers (81% have a validated US residence), the fast-course premium falls from 9.7 to 6.7 percentage points and the qualifying-venue multiple over the majors from 2.6 to 1.9. An external check against BAA-published qualifying-race counts for Boston 2025 shows similar recovery across course types (84% at Chicago versus 80% at CIM). The 2026 comparison is retained as a flagged sensitivity rather than the trusted endpoint because the broader course-level recovery spread is larger.

The direct participant line complements, rather than forms an additional additive step in, the random-assignment chain. Figure E4 compares the fast-course share among effective qualifiers with the share among linked Boston entrants; the remaining gap combines genuine participation selection and any residual differential linkage.

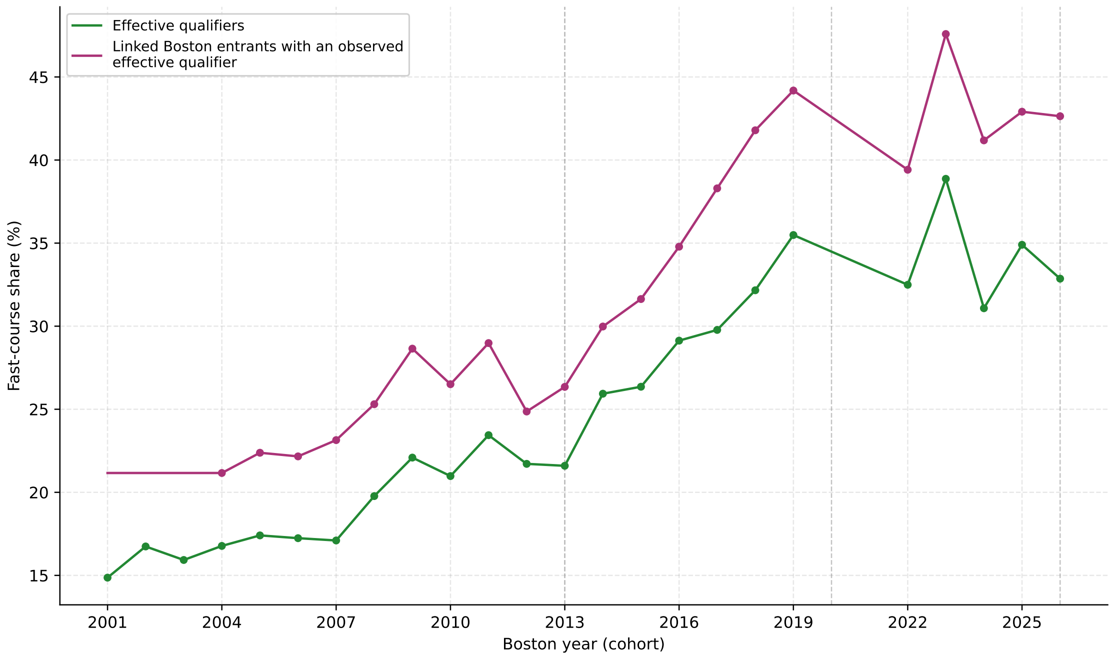

Figure E4: Fast-course exposure among effective qualifiers and among linked Boston entrants with an observed effective qualifier. Each participant is counted once using the fastest observed performance clearing the effective cutoff. COVID-era cohorts omitted. The gap between the two series combines participation selection and residual linkage differences; it is not a pure measure of BQ take-up.

E.5 Courses That Attract Marginal Runners

A course-level diagnostic identifies which courses attract the marginal runners. For each course-cohort cell, I compute the share of finishers within ±3% of their standard and benchmark it against a smooth in course speed: the residual is the course's over- or under-attraction of marginal runners beyond what its speed predicts. The top over-attractors are the "last chance" BQ events—Last Chance BQ.2 Grand Rapids, Last Chance BQ.2 Chicagoland, and Spring Chance BQ.2—whose marginal-runner shares sit roughly double what course speed alone would predict (Figure E5).

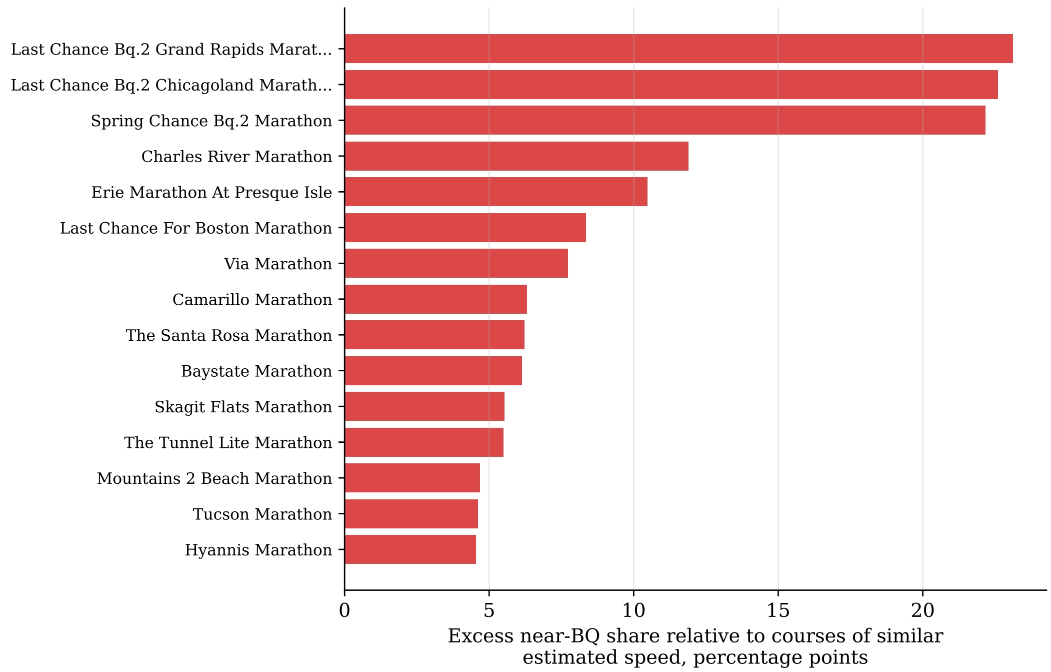

Figure E5: Courses that disproportionately attract near-BQ runners, ranked by the cohort-weighted excess near-BQ share relative to courses of similar estimated speed. Dedicated BQ products lead the over-attractors.

E.6 Selection and Excess-Mass Tables

Table E1: Selection on predictable pre-race state: pressure, venue speed, and the announcement difference-in-differences.

	(1)	(2)	(3)	(4)	(5)
Dep. variable:	Course speed (log FE)	BQ pressure (index 0–1)	Fast course (pp)	Qualifies (pp)	Boston take-up (pp)
BQ pressure	-0.0151***		11.18***	24.70***	28.02***
	(0.0002)		(0.18)	(0.26)	(0.28)
Chosen-course speed (log FE)		-0.5773***			
		(0.0063)			
Near BQ (±5 min)	0.0058***		-5.26***	-9.35***	-5.69***
	(0.0002)		(0.18)	(0.24)	(0.30)
Near BQ × post-2011	-0.0022***		3.16***	0.00	-4.31***
	(0.0002)		(0.16)	(0.20)	(0.25)
Post-2011	-0.0047***	-0.0062***	5.83***	-1.33***	-1.52***
	(0.0001)	(0.0007)	(0.05)	(0.09)	(0.08)
Log distance to course	-0.0002***	0.0038***	0.71***	0.36***	0.49***
	(0.0000)	(0.0001)	(0.01)	(0.01)	(0.01)
Age, gender, race-count controls	Yes	Yes	Yes	Yes	Yes
Runner-years	4,379,601	4,379,601	4,379,601	4,379,601	4,379,601

Notes: OLS on runner-year observations of geocoded US-resident runners with at least one prior observed race. Each column's dependent variable is named in the header: columns (3)–(5) are linear probability models with coefficients and standard errors displayed in percentage points; column (1) is the chosen course's log FE and column (2) the pressure index. BQ pressure is a tent-shaped index of the runner's prior-best margin, peaked eight minutes outside the standard. Standard errors clustered by runner in parentheses. The event-study version of the near-BQ × post-2011 contrast is Figure 4, Panel B. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$.

Table E2: Polynomial excess-mass estimates at the BQ threshold and round-number placebo, by course type and era.

	Pre-2012		2012+	
	Excess mass	N	Excess mass	N
Excess mass at the BQ threshold				
All courses	2.19	1,182,288	1.23	1,360,857
	[2.12, 2.26]		[1.18, 1.29]	
Fast courses	3.23	170,632	1.65	328,675
	[3.03, 3.43]		[1.54, 1.76]	
Medium courses	2.14	758,610	1.08	846,875
	[2.06, 2.21]		[1.00, 1.16]	
Slow courses	1.49	253,046	1.00	185,307
	[1.34, 1.65]		[0.83, 1.18]	
Excess mass at the 4:00 round-number placebo				
All courses	1.83	1,726,393	1.96	2,015,954
	[1.77, 1.88]		[1.91, 2.01]	
Fast courses	2.05	190,094	1.71	352,471
	[1.86, 2.22]		[1.60, 1.83]	
Medium courses	1.85	1,095,679	2.08	1,282,476
	[1.79, 1.92]		[2.01, 2.14]	
Slow courses	1.67	440,620	1.80	381,007
	[1.56, 1.79]		[1.68, 1.92]	

Notes: Excess mass is the standard normalized excess-mass estimator (Saez, 2010; Kleven and Waseem, 2013): the density piled into the bunching interval $[0, 3)$ minutes inside the threshold, relative to a degree-6 polynomial counterfactual fit to the binned density (bin width 0.5 minutes) over ± 30 minutes around the threshold excluding the window $[-2, +5]$ minutes. Brackets are 95% Poisson bin-resample bootstrap intervals (300 draws); N is the cell sample. The placebo panel recenters on 4:00 and excludes runners for whom 240 minutes is a BQ standard. Round-number bunching is present everywhere; the identifying contrast is the fast$-$slow gradient, roughly four times larger at the BQ threshold than at the placebo before 2012, and present only at the BQ threshold after 2012.

Table E3: Reconciling the sorting magnitudes: each estimate, its estimand, its population, and the channels it includes.

Estimate	Value	Estimand	Population	Channels
Excess fast-course share, near-BQ	+7.9 pp	Stock: fast-course share of near-BQ finishers minus the era's full field	Near-BQ finishers (neutral ±5 min), 2013–2019	All BQ-related sorting (venue choice + correlated effort)
Announcement difference-in-differences	+3.1 pp (SE 0.2)	Change: differential fast-course choice of near-BQ runners after the 2011 tightening, net of year and runner controls	US runners with geocoded home	Venue choice (demand) response to the announcement
Misallocation rate (screen side)	9% → 17%	Stock: share of apparent qualifiers whose qualification depends on the course they ran (2004 → 2025 cohorts)	Apparent qualifiers at indexed courses	Outcome of all channels combined, screen side

Appendix F Policy Implementation and Robustness

This appendix records the implementation details behind Section 4: the exact coding of each rule, each rule's coverage and precision against the common-course benchmark, and the course-level reallocation.

The rules are coded as follows. The BAA rule adds five minutes to qualifying times from courses with 1,500–2,999 feet of net descent and ten minutes for 3,000–5,999 feet, and rejects times from courses dropping 6,000 feet or more; rejected performances are treated as unable to qualify at any margin. The symmetric elevation formula adjusts every profiled course by its predicted speed from the physical-and-weather regression of Table B4 (net drop, gross climbing, maximum elevation, and typical weather), so fast courses receive penalties and slow courses receive credits; unprofiled courses are left unadjusted. The course-effect benchmark adjusts every indexed course by the behavior-adjusted index itself. All rules are evaluated on the same fixed field of US 2024–25 qualifying performances, scored against the common-course benchmark of Section 4.

Table F1: Policy rules against the common-course benchmark: coverage, turnover, and precision.

Rule	Courses adj. (%)	Perf. adj. (%)	Field chg. (%)	Disagr. w/ bench. (%)	Gap closed (%)	Removal prec. (%)	Addition prec. (%)
Common cutoff	0.0	0.0	0.0	11.9	0.0	–	–
BAA 2027 net-drop	3.6	3.1	1.9	11.1	6.3	86.5	53.3
Symmetric elevation	86.3	98.5	6.7	9.2	22.5	72.4	67.8
Course-effect benchmark	100.0	100.0	11.9	0.0	100.0	100.0	100.0

Notes: US road qualifying market, 2024–25 performances, fixed field 87,066. “Courses adjusted” and “performances adjusted” are the shares each rule touches. “Field changed” is the rule’s one-sided turnover relative to the common cutoff. “Disagreement with benchmark” is the share of the field differing from the common-course benchmark’s qualifier set, $|S_r \triangle S^*|/2N$, and “gap closed” is the share of the common-cutoff disagreement the rule eliminates. Removal (addition) precision is the share of the performances a rule removes (promotes) that the benchmark also removes (promotes). Scoring against a leave-out re-estimate of the course effects, or against any index in the corrected suite, changes the BAA rule’s captured share by at most a few percentage points (4.8–7.6%).

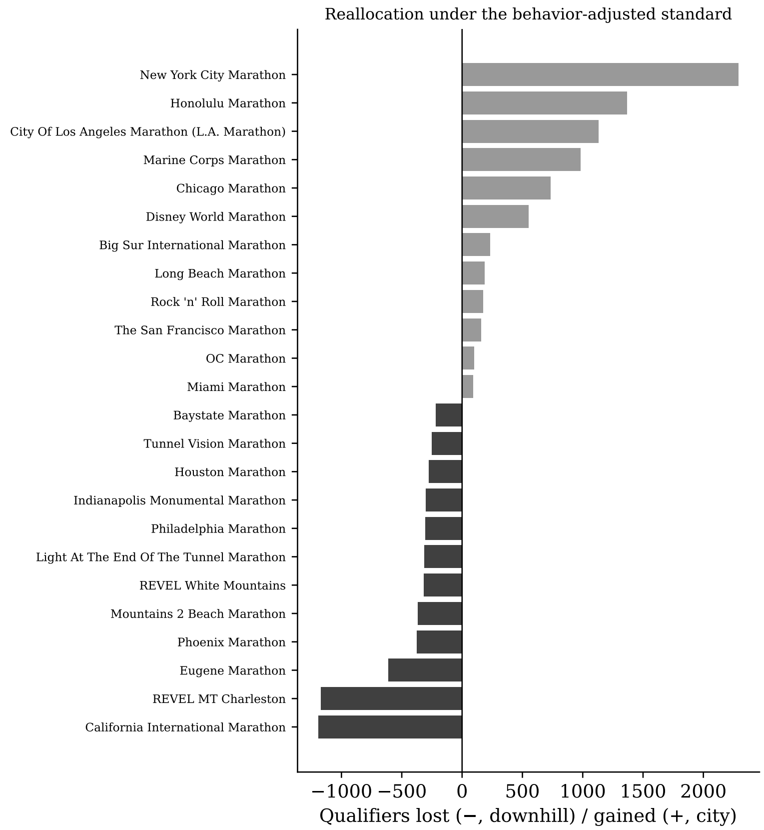

Figure F1: Courses losing and gaining qualifiers under the common-course benchmark, fixed field. Negative bars are qualifiers lost (concentrated at downhill and purpose-built qualifying courses); positive bars are qualifiers gained (concentrated at the large city marathons slower than Boston).